

Spatio-Temporal Statistical Models

Nearby values tend to be more alike than those far apart; this chapter explores this notion, where “nearby” refers to being proximal in both space and time. Let $\{Y(\mathbf{s}; t) : \mathbf{s} \in D_s, t \in D_t\}$ denote a spatio-temporal random process that is a statistical model for a phenomenon evolving through the spatio-temporal index set $D_s \times D_t$. For example, $Y(\mathbf{s}; t)$ might be the air pressure at time t and at geographical coordinates $\mathbf{s} = (\text{latitude}, \text{longitude}, \text{height})'$; conceptually, $D_s = \text{sphere} \times \mathbb{R}$ and $D_t = \mathbb{R}$. Or, $Y(\mathbf{s}; t)$ might be the yearly breast-cancer incidence rate in each of the 88 counties of Ohio; conceptually, $D_s = \{\mathbf{s}_1, \dots, \mathbf{s}_{88}\}$, the set of centroids of each county, and D_t is a subset of the positive integers (denoting years). There may also be instances where the spatio-temporal index set should be written more generally as $D_{s,t}$, to account for temporally evolving spatial index sets or curved reference frames (see Einsteinian physics in Chapter 1).

In this book, we emphasize *dynamical* spatio-temporal statistical models, because we believe that the holy grail of Science, namely the discovery of causation, is within our grasp when we model how current process values have evolved from past process values. In cases where scientific understanding is not available to develop a dynamical spatial statistical model, a *descriptive* statistical model based on first- and second-order moments (means, variances, and covariances) can be developed. Because the dynamical models typically yield first- and second-order moments, the descriptive models are mathematically more general but are scientifically less interpretable.

The dynamical statistical approach is built from *conditional* probability distributions, modeling how the current value $Y(\mathbf{s}; t)$ behaves given the behavior of “nearby” current and past values taken from the set $\{Y(\mathbf{x}; r) : \mathbf{x} \in D_s, r < t\} \cup \{Y(\mathbf{x}; t) : \mathbf{x} \neq \mathbf{s}\}$. These conditional-probability specifications come from scientific knowledge. In the absence of adequate knowledge, the data can still provide information about the spatio-temporal variability, and this is typically achieved through *descriptive* models based on *marginal* probability distributions. An example of this difference in approach to statistical modeling

occurs in linear mixed models, where one chooses to either (a) make explicit the random components through conditional probability distributions or (b) make explicit the mean vector and covariance matrix through marginal probability distributions.

Since marginal probabilities can be obtained from weighted integrals of conditional probabilities, models based on conditional probabilities are closer to the etiology of the phenomenon under study. Indeed, different physical processes could imply the same marginal probabilities; see the example given at the beginning of Section 4.2, where a symmetric spatial process is matched to a unilateral spatial process. That is, the etiology of the phenomenon under study can be masked by using a first- and second-order-moment description. Nevertheless, descriptive spatio-temporal statistical models based on marginal probability distributions do have a place when the modeler's knowledge is limited. Otherwise, our preference is to build dynamical spatio-temporal models based on conditional probability distributions.

Physical Statistical Models

When there is scientific knowledge about a process Y , it can often be described in terms of a partial differential equation (PDE), sometimes in time, sometimes in space, and sometimes in both. Heine (1955) considers such equations in two dimensions—in particular, the general second-order, linear, nonstochastic PDE:

$$\left\{ a \frac{\partial^2}{\partial u^2} + 2h \frac{\partial^2}{\partial u \partial v} + b \frac{\partial^2}{\partial v^2} + 2g \frac{\partial}{\partial u} + 2f \frac{\partial}{\partial v} + c \right\} Y(u, v) = 0.$$

He shows that by simple changes of variables, this PDE reduces to several special cases.

$$\text{Parabolic:} \quad \left\{ \left(\frac{\partial}{\partial u} + \alpha \right)^2 - \gamma^2 \left(\frac{\partial}{\partial v} \pm \beta \right) \right\} Y(u, v) = 0,$$

$$\text{Elliptic:} \quad \left\{ \left(\frac{\partial}{\partial u} - \alpha \right)^2 + \frac{\partial^2}{\partial v^2} \pm \gamma^2 \right\} Y(u, v) = 0,$$

$$\text{Hyperbolic:} \quad \left\{ \left(\frac{\partial}{\partial u} + \alpha \right) \left(\frac{\partial}{\partial y} + \beta \right) \pm \gamma^2 \right\} Y(u, v) = 0,$$

$$\text{Degenerate:} \quad \left\{ \gamma \frac{\partial^2}{\partial v^2} + \alpha \frac{\partial}{\partial v} \pm \beta \right\} Y(u, v) = 0,$$

where α , β , and γ are generic, nonnegative parameters.

A nonstochastic (or deterministic) PDE can be converted into a *stochastic* PDE by replacing 0 on the PDE's right-hand side with a (mean-zero) stochastic term. In a remarkable early contribution to the Statistics literature, Hotelling (1927) investigated a number of one-dimensional (in time) stochastic PDEs.

Heine (1955) did the same thing in two dimensions, based on the equations given above; here, we write the stochastic term in the PDE as $\delta(u, v)$, where $\delta(\cdot, \cdot)$ is a mean-zero, two-dimensional stochastic process that models either (a) random impulses from smaller-order contributions or (b) uncertainty in the scientific relationship expressed through the PDE. The resulting equation is an example of a *physical statistical model* (Berliner, 2003). Clearly, the process $Y(\cdot, \cdot)$ is now a stochastic process, and its moments are of interest. Because $E(\delta(u, v)) \equiv 0$, linearity of the PDE implies that $E(Y(u, v)) \equiv 0$. What is $\text{cov}(Y(u, v), Y(p, q))$?

The two-dimensional co-ordinates (u, v) could be purely spatial or they could be spatio-temporal. We consider the case of a spatial u and a temporal v , and we define the spatio-temporal process $\{Y(s; t) : s \in D_s \subset \mathbb{R}, t \in D_t \subset \mathbb{R}\}$ according to a stochastic PDE that is, for illustration, parabolic. In particular, we consider the spatio-temporal process presented by Heine (1955) and Jones and Zhang (1997), which is governed by the following *stochastic* PDE:

$$\frac{\partial Y(s; t)}{\partial t} - \beta \frac{\partial^2 Y(s; t)}{\partial s^2} + \alpha Y(s; t) = \delta(s; t), \quad (6.1)$$

where $\alpha > 0$, $\beta > 0$, and $\delta(\cdot; \cdot)$ is an *error process* that is random and has mean zero. Clearly, a $Y(\cdot; \cdot)$ that satisfies (6.1) is an example of a dynamical spatio-temporal statistical model.

Whittle (1963) considers the same class, but where s is more generally a spatial location in $\mathbb{R}^d$ (see also Section 6.3.1). These models were called *diffusion-injection models* by Whittle (1986, p. 431), and we shall adopt that terminology here (“injection” here refers to injection of variability from the random component δ). In all that follows in this chapter, $\delta(\cdot; \cdot)$ is assumed to be a spatio-temporal process with mean zero, independent in time, and potentially dependent in space (at any fixed time). The special case of independence in space corresponds to the covariance structure, $\text{cov}(\delta(s; t), \delta(x; r)) = \sigma_\delta^2 I(s = x, t = r)$, which is white noise in $\mathbb{R}^1$.

The *deterministic* PDE associated with (6.1) is, using obvious shorthand,

$$\frac{\partial Y}{\partial t} = \beta \frac{\partial^2 Y}{\partial s^2} - \alpha Y, \quad (6.2)$$

which is a homogeneous reaction-diffusion model (homogeneous because every term contains the dependent variable, Y , or its derivatives). In (6.2), it is understood that the partial derivatives are evaluated at s and t . The equation specifies that the rate of change in Y is equal to the “spread” of Y in space (diffusion) offset by the “loss” of a certain multiple of Y (reaction). From a given initial condition, $Y(s; 0)$, the process $Y(\cdot; \cdot)$ governed by the deterministic PDE (6.2) dampens to zero as time t increases, which is not a very realistic behavior for many dynamical processes seen in nature (e.g., meteorological processes). Figure 6.1 shows several plots of $Y(s; t)$ defined by (6.2), as a function of space s and time t , for different choices of α and β . (The solutions shown in Figure 6.1 are obtained numerically.)

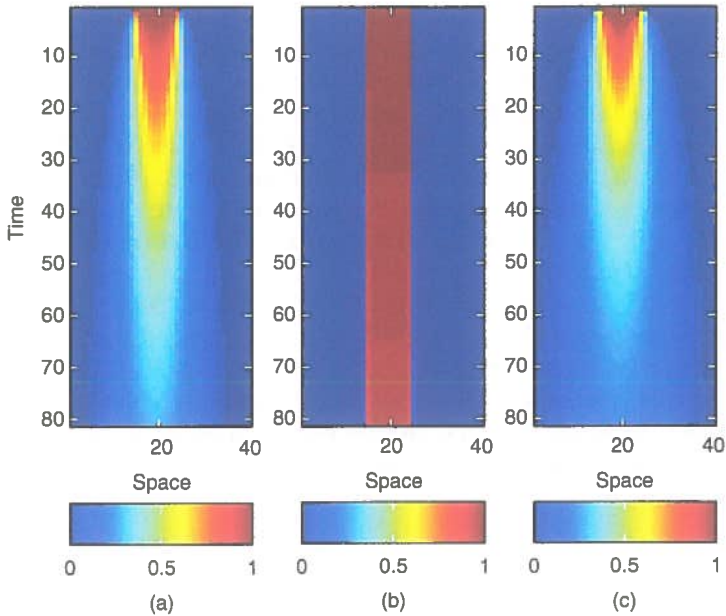


Figure 6.1 Several plots of $Y(s; t)$ defined by the deterministic reaction-diffusion equation described in (6.2), as a function of space s and time t for different choices of α and β : (a) $\alpha = 1$, $\beta = 20$; (b) $\alpha = 0.05$, $\beta = 0.05$; (c) $\alpha = 1$, $\beta = 50$. The starting values are $Y(s; 0) = 1$ for $15 \leq s \leq 24$, and 0 otherwise. (The solutions are obtained numerically for $s \in D_s \equiv \{\Delta_s, 2\Delta_s, \dots, 40\Delta_s\}$, where $\Delta_s = 1$, and $t \in D_t \equiv \{\Delta_t, 2\Delta_t, \dots, 80\Delta_t\}$, where $\Delta_t = 0.1$.)

By including the random process $\delta(\cdot; \cdot)$ in (6.1), a statistical balance is reached between (a) the “disturbance” caused by $\delta(\cdot; \cdot)$ and (b) the smoothing effect of both the diffusion component and the loss component. That is, from a given initial condition, the stochastic PDE results in a process that eventually achieves both spatial and temporal stationarity. Figure 6.2 shows several simulations of $Y(s; t)$ defined by the stochastic PDE (6.1), for $\alpha = 1$, $\beta = 20$, δ is white noise, and different choices of its variance σ_δ^2 ; these should be compared to Figure 6.1a, the (numerical) solution of the corresponding deterministic PDE.

Typically, stochastic PDEs like those considered by Heine (1955) are difficult to deal with analytically. In the case of the stochastic PDE (6.1), Heine (1955) shows that $Y(\cdot; \cdot)$ has a stationary covariance function $C_Y(\cdot; \cdot)$, and he derives an analytical expression for the corresponding spatio-temporal correlation function: For $h \in \mathbb{R}$ and $\tau \in \mathbb{R}$, we obtain

$$\begin{aligned}
 C_Y(h; \tau) / C_Y(0; 0) &\equiv \rho_Y(h; \tau) \\
 &= (1/2) \left\{ e^{-h(\alpha/\beta)^{1/2}} \operatorname{Erfc} \left(\frac{2\tau(\alpha/\beta)^{1/2} - h/\beta}{2(\tau/\beta)^{1/2}} \right) \right. \\
 &\quad \left. + e^{h(\alpha/\beta)^{1/2}} \operatorname{Erfc} \left(\frac{2\tau(\alpha/\beta)^{1/2} + h/\beta}{2(\tau/\beta)^{1/2}} \right) \right\}, \quad (6.3)
 \end{aligned}$$

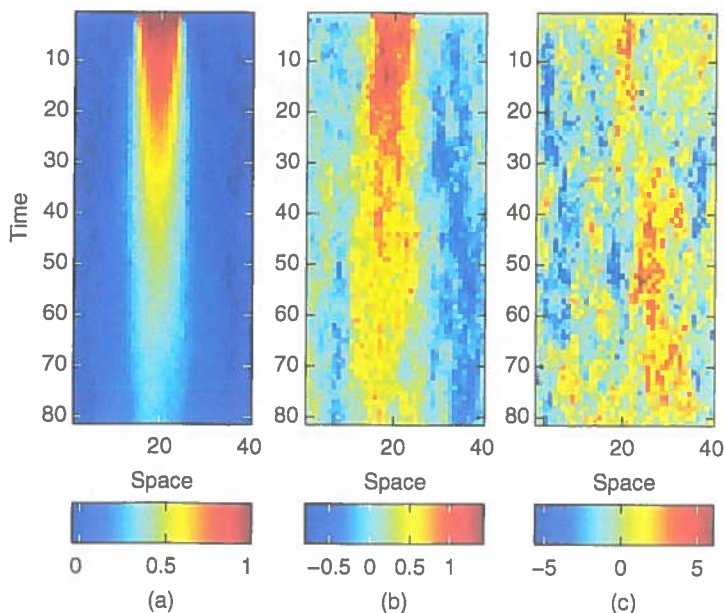


Figure 6.2 Simulations of $Y(s; t)$ defined by the stochastic PDE (6.1) as a function of space $s \in D_s$ and time $t \in D_t$, and δ is mean-zero white noise with variance σ_δ^2 . The same $\alpha = 1$, $\beta = 20$, and starting values, $Y(s; 0)$, as in Figure 6.1a are used: (a) $\sigma_\delta = 0.01$, (b) $\sigma_\delta = 0.1$, and (c) $\sigma_\delta = 1$.

where $\text{Erfc}(z)$ is the “complementary error function,”

$$\text{Erfc}(z) \equiv (2/\pi^{1/2}) \int_z^\infty e^{-v^2} dv, \quad z \geq 0,$$

and for $z < 0$, $\text{Erfc}(z) = 2 - \text{Erfc}(-z)$. Figure 6.3 shows a contour plot of $\rho_Y(h; \tau)$ given by (6.3) as a function of spatial lag h and temporal lag τ , for $\alpha = 1$ and $\beta = 20$ (the same parameter values used in Figure 6.1a and Figure 6.2).

Special cases include the spatial correlation function at a given time:

$$\rho_Y(h; 0) = e^{-h(\alpha/\beta)^{1/2}}, \quad h > 0, \quad (6.4)$$

and the temporal correlation function at a given spatial location:

$$\rho_Y(0; \tau) = \text{Erfc}(\tau^{1/2} \alpha^{1/2}), \quad \tau > 0. \quad (6.5)$$

Figure 6.4a shows a plot of $\rho_Y(h; 0)$ given by (6.4) as a function of spatial lag h , and Figure 6.4b shows a plot of $\rho_Y(0; \tau)$ given by (6.5) as a function of temporal lag τ ; both figures are for $\alpha = 1$ and $\beta = 20$, as in Figure 6.2.

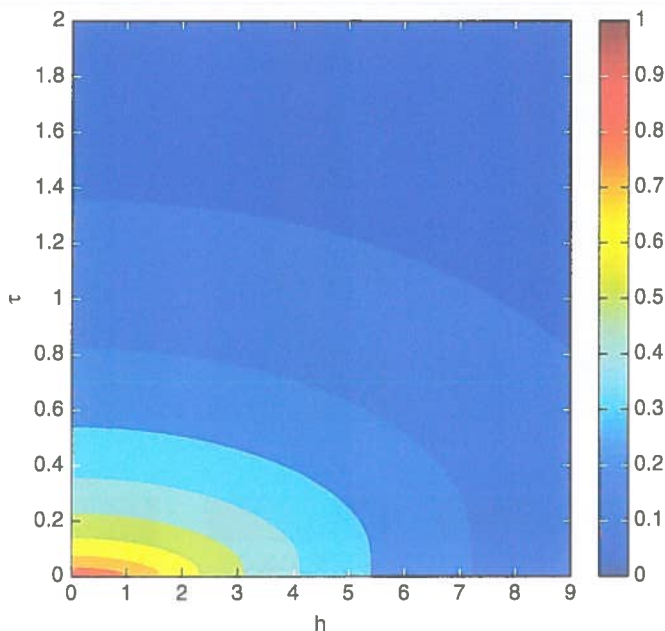


Figure 6.3 Contour plot of $\rho_Y(h; \tau)$ defined in (6.3) as a function of spatial lag h and temporal lag τ ; $\alpha = 1$ and $\beta = 20$.

The stochastic PDEs considered above are second-order linear. Whittle (1963) discussed the more general case of

$$\mathcal{L}Y(\mathbf{s}; t) = \delta(\mathbf{s}; t),$$

where $\mathcal{L}$ is a linear operator. The spatial version of this is $\mathcal{L}Y(\mathbf{s}) = \delta(\mathbf{s})$; $\mathbf{s} \in D_s \subset \mathbb{R}^d$. He gives the example

$$\left\{ \frac{\partial^2}{\partial s_1^2} + \cdots + \frac{\partial^2}{\partial s_d^2} - \alpha^2 \right\}^p Y(\mathbf{s}) = \delta(\mathbf{s}), \quad (6.6)$$

where $\mathbf{s} = (s_1, \dots, s_d)'$, $\delta(\cdot)$ is a mean-zero, white-noise process in $\mathbb{R}^d$ (i.e., $\text{cov}(\delta(\mathbf{s}), \delta(\mathbf{x})) = \sigma_\delta^2 I(\mathbf{s} = \mathbf{x})$), and the linear operator on the left-hand side is a fractional Laplacian (of fraction p) defined through its Fourier representation (e.g., Fuentes, Chen, and Davis, 2008). Upon reparameterizing, $\theta_1 = 1/\alpha > 0$, $\theta_2 = 2p - d/2 > 0$ (i.e., restrict $p > d/4$), Whittle (1963) shows that the (spatial, stationary, isotropic) covariance function of $Y(\cdot)$ defined by (6.6) is given by

$$C_Y(\mathbf{h}) \propto \{\|\mathbf{h}\|/\theta_1\}^{\theta_2} K_{\theta_2}(\|\mathbf{h}\|/\theta_1), \quad (6.7)$$

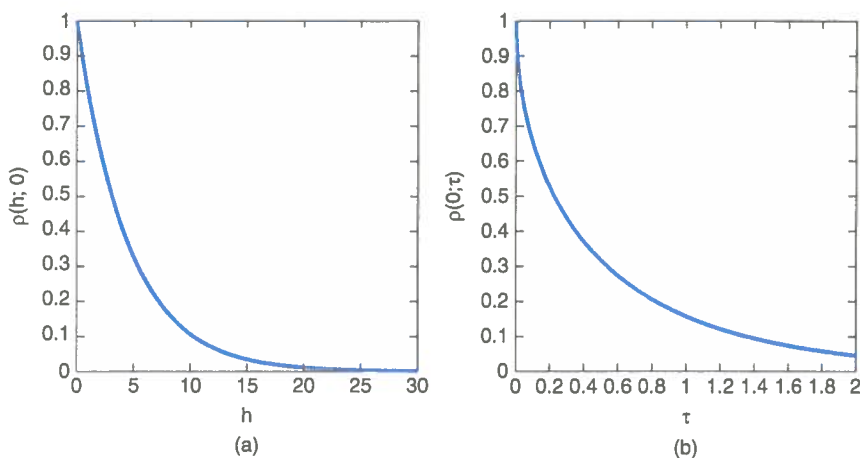


Figure 6.4 (a) Plot of marginal spatial correlation $\rho_Y(h; 0)$, given in (6.4), as a function of spatial lag h ; $\alpha = 1$ and $\beta = 20$. (b) Plot of marginal temporal correlation $\rho_Y(0; \tau)$, given in (6.5), as a function of temporal lag τ ; $\alpha = 1$ and $\beta = 20$.

where K_{θ_2} is a modified Bessel function of the second kind of order θ_2 (e.g., Abramowitz and Stegun, 1964, pp. 374–379). The covariance function (6.7) is known as the *Matérn covariance function*; see (4.6).

We now discuss some initial data analysis that addresses the question of whether there is spatio-temporal dependence in the dataset under study. Griffith (1988) considered the Space–Time Index (STI) as a type of Moran’s I statistic for the spatio-temporal setting. Suppose that there are m spatial locations $\mathbf{s}_1, \dots, \mathbf{s}_m$, at each of which there are T mean-corrected observations, $\{Z(\mathbf{s}_i; t) : i = 1, \dots, m; t = 1, \dots, T\}$. Henebry (1995) discussed the properties of

$$\text{STI} \equiv \frac{m(T-1) \sum_{t=2}^T \sum_{j=1}^m \sum_{i=1}^m w_{ij,t-1} Z(\mathbf{s}_i; t) Z(\mathbf{s}_j; t-1)}{(\sum_{t=2}^T \sum_{j=1}^m \sum_{i=1}^m w_{ij,t-1})(\sum_{t=1}^T \sum_{i=1}^m Z(\mathbf{s}_i; t)^2)},$$

where $\{w_{ij,t}\}$ are spatio-temporal weights indicating spatial “neighborliness” at each snapshot of time.

The permutation distribution of STI (i.e., permute the locations $\{\mathbf{s}_i : i = 1, \dots, m\}$ and the time points $\{1, 2, \dots, T\}$, and recompute STI for each permutation) serves as a baseline for the variability of STI under the null hypothesis (H_0) of *no* spatio-temporal dependence. An initial check to see whether it is worthwhile fitting a spatio-temporal model to data could be carried out as follows: Generate the permutation distribution of STI, which is very computationally intensive. Observe where the actual *observed* STI falls in the distribution generated. If it is below the $100(\alpha/2)$ th percentile or above the $100(1 - \alpha/2)$ th percentile, then reject H_0 at the $100(1 - \alpha)\%$ significance level. Practically speaking, a rejection of H_0 means that the spatio-temporal dependence is present and a spatio-temporal model is needed.

Space and time can be continuous or discrete, depending on the context. Two motivating examples were given at the beginning of this chapter: In the first example, $\mathbf{s} = (\text{latitude}, \text{longitude}, \text{height})'$, which is a continuous spatial index; and in the second example, $\mathbf{s} = \text{centroid of a county in Ohio}$, which is a discrete spatial index. For the spatial dimension, discreteness is often (although not always) caused by aggregation of the continuous index into regions. For the temporal dimension, discreteness is often (although not always) caused by sampling the continuous index at equally spaced time intervals. Furthermore, it could be that data are collected based on one type of index but inference is desired for the other type. For example, the problem of downscaling comes from having data for a given, coarse spatial resolution, from which one wishes to infer the underlying process at a much finer spatial resolution (i.e., change-of-support; Chapter 1, Section 4.1.3, Section 7.1.2). Suppose we cross-classify the two types of index (continuous/discrete) with the two types of dimension (spatial/temporal), which results in four categories of spatio-temporal models. Where appropriate, we shall mention the category to which a given model belongs. For the temporal index, we use a notational prop: When time is continuous, an element of the spatio-temporal process is written as $Y(\mathbf{s}; t)$, and when time is discrete, an element of the spatio-temporal process is written as $Y_t(\mathbf{s})$.

Spatio-temporal covariance functions offer a succinct but informative summary of random processes on $D_s \times D_t$. Indeed, it is very common to consider the covariance function as a characterization of the spatio-temporal process $Y(\cdot; \cdot)$. An example given in Chapter 1 for Gaussian processes shows that modeling based purely on the covariance function is ambiguous with respect to the model's dynamics. In this chapter, we discuss spatio-temporal covariance modeling (based on marginal probabilities), along with dynamical modeling (based on conditional probabilities). It is worth reemphasizing that, when possible, we prefer to work with dynamical models directly, where the etiology of the phenomenon is at its most apparent and causative effects can be incorporated. In that sense, this chapter is a bridge between the exploratory methods presented in Chapter 5 and the dynamical spatio-temporal HMs presented in Chapter 7.

6.1 SPATIO-TEMPORAL COVARIANCE FUNCTIONS

To motivate spatio-temporal covariance functions in cases where the dynamical structure is unknown, write the spatio-temporal process $Y(\cdot; \cdot)$ as

$$Y(\mathbf{s}; t) \equiv \mu(\mathbf{s}; t) + \beta(\mathbf{s}) + \gamma(t) + \kappa(\mathbf{s}; t) + \delta(\mathbf{s}; t), \quad \mathbf{s} \in D_s, \quad t \in D_t, \quad (6.8)$$

which is similar to the mixed-effects model discussed by Wikle (2003a). In (6.8), $\mu(\mathbf{s}; t)$ is a *deterministic* mean; $\beta(\mathbf{s})$ is a mean-zero *random* effect representing location-specific variability common to all times; $\gamma(t)$ is a mean-zero

random effect representing time-specific variability common to all locations; $\kappa(\mathbf{s}; t)$ is a mean-zero *random* effect capturing the spatio-temporal interaction not found in the larger-scale deterministic $\mu(\mathbf{s}; t)$; $\delta(\mathbf{s}; t)$ is a mean-zero *random* effect representing the microscale spatio-temporal variability that is modeled by white noise in $D_s \times D_t$; and all random effects are mutually statistically independent. To make the exposition simple, we present (6.8) as a continuous-space, continuous-time model. Further, the second-order moments are written as

$$\text{cov}(\beta(\mathbf{s}), \beta(\mathbf{x})) \equiv C_\beta(\mathbf{s}, \mathbf{x}) \quad (6.9)$$

$$\text{cov}(\gamma(t), \gamma(r)) \equiv C_\gamma(t, r) \quad (6.10)$$

$$\text{cov}(\kappa(\mathbf{s}; t), \kappa(\mathbf{x}; r)) \equiv C_\kappa(\mathbf{s}, \mathbf{x}; t, r) \quad (6.11)$$

$$\text{var}(\delta(\mathbf{s}; t)) \equiv \sigma_\delta^2. \quad (6.12)$$

Covariance models for (6.9) can be found in Chapter 4, covariance models for (6.10) can be found in Chapter 3, and covariance models for (6.11) are discussed at length in this section.

We now present some examples of (6.8), where, for simplicity, we assume that the deterministic mean $\mu(\mathbf{s}; t) \equiv 0$. Using marginal probabilities, we calculate the resulting spatio-temporal covariance functions, although we note that there is a conditional-probability interpretation of (6.8); see Section 6.6.

Regional-Effects Model (Reinsel et al., 1981; Bloomfield et al., 1983)

Suppose that $Y(\cdot; \cdot)$ can be written as

$$Y(\mathbf{s}; t) = \gamma(t) + \kappa_k(t) + \delta(\mathbf{s}; t), \quad \mathbf{s} \in D_s^{(k)}, \quad (6.13)$$

where D_s is made up of mutually exclusive regions $\{D_s^{(k)} : k = 1, \dots, K\}$. The random effect, $\kappa_k(t)$, is a “regional” component of variation *common* to all locations $\mathbf{s}$ in the k th region $D_s^{(k)}$; they are independent for $k = 1, \dots, K$. Suppose that all random effects are white noise:

$$\text{var}(\gamma(t)) \equiv \sigma_\gamma^2, \quad \text{var}(\kappa_k(t)) \equiv \sigma_\kappa^2, \quad \text{var}(\delta(\mathbf{s}; t)) = \sigma_\delta^2. \quad (6.14)$$

This model results in temporal independence but induces “block” spatial covariances. For all t , we have

$$\text{cov}(Y(\mathbf{s}; t), Y(\mathbf{x}; t)) = \begin{cases} \sigma_\gamma^2 + \sigma_\kappa^2 + \sigma_\delta^2, & \text{if } \mathbf{s} = \mathbf{x} \\ \sigma_\gamma^2 + \sigma_\kappa^2, & \text{if } \mathbf{s}, \mathbf{x} \in D_s^{(k)}, \mathbf{s} \neq \mathbf{x}; \\ & k = 1, \dots, K \\ \sigma_\gamma^2, & \text{otherwise.} \end{cases} \quad (6.15)$$

The contribution of Bloomfield et al. (1983) was to augment the spatio-temporal random effect, $\kappa_k(t)$ in (6.13), with another spatio-temporal random effect $\eta(\cdot; \cdot)$ that exhibits temporal correlation. Suppose that

$$Y(\mathbf{s}; t) \equiv \gamma(t) + \kappa_k(t) + \eta(\mathbf{s}; t) + \delta(\mathbf{s}; t), \quad (6.16)$$

where for all t, r :

$$\text{cov}(\eta(\mathbf{s}; t), \eta(\mathbf{x}; r)) = \begin{cases} \sigma_\eta^2, & \text{if } \mathbf{s} = \mathbf{x}, \\ C_\eta(\mathbf{s}, \mathbf{x}), & \text{if } \mathbf{s} \neq \mathbf{x}. \end{cases} \quad (6.17)$$

Then (6.15) is generalized to become

$$\text{cov}(Y(\mathbf{s}; t), Y(\mathbf{x}; r)) = \begin{cases} \sigma_\gamma^2 + \sigma_\kappa^2 + \sigma_\eta^2 + \sigma_\delta^2, & \text{if } \mathbf{s} = \mathbf{x}, \\ \sigma_\gamma^2 + \sigma_\kappa^2 + C_\eta(\mathbf{s}, \mathbf{x}), & \text{if } \mathbf{s}, \mathbf{x} \in D_s^{(k)}, \mathbf{s} \neq \mathbf{x}; \\ & k = 1, \dots, K, \\ \sigma_\gamma^2 + C_\eta(\mathbf{s}, \mathbf{x}), & \text{otherwise.} \end{cases} \quad (6.18)$$

Mixture Model (Oehlert, 1993)

Consider the expression (6.8), with (6.9) given by $C_\beta(\mathbf{s}, \mathbf{x}) = \sigma_\beta^2 I(\mathbf{s} = \mathbf{x})$, and (6.11) given by a separable covariance, $C_\kappa(\mathbf{s}, \mathbf{x}; t, r) = C_\kappa^{(s)}(\mathbf{s}, \mathbf{x}) \cdot C_\kappa^{(t)}(t, r)$. Then

$$\begin{aligned} & \text{cov}(Y(\mathbf{s}; t), Y(\mathbf{x}; r)) \\ &= \begin{cases} \sigma_\beta^2 + C_\gamma(t, t) + C_\kappa^{(s)}(\mathbf{s}, \mathbf{s}) \cdot C_\kappa^{(t)}(t, t) + \sigma_\delta^2, & \text{if } \mathbf{s} = \mathbf{x}, t = r, \\ C_\gamma(t, r) + C_\kappa^{(s)}(\mathbf{s}, \mathbf{x}) \cdot C_\kappa^{(t)}(t, r), & \text{otherwise.} \end{cases} \end{aligned} \quad (6.19)$$

When C_γ , $C_\kappa^{(s)}$, and $C_\kappa^{(t)}$ in (6.19) are stationary (in space or time), we obtain the type of mixture models proposed by De Iaco, Myers, and Posa (2001); see (6.30) below.

Invariant-Temporal-Covariance Model (Stein, 1986)

Consider (6.8) with $\gamma(t) \equiv 0$, and (6.11) given by the temporally invariant covariance function, $C_\kappa(\mathbf{s}, \mathbf{x}; t, r) \equiv C_\kappa^{(s)}(\mathbf{s}, \mathbf{x})$, which is exactly the same model form as (6.17). Then

$$\text{cov}(Y(\mathbf{s}; t), Y(\mathbf{x}; r)) = \begin{cases} C_\beta(\mathbf{s}, \mathbf{s}) + C_\kappa^{(s)}(\mathbf{s}, \mathbf{s}) + \sigma_\delta^2, & \text{if } \mathbf{s} = \mathbf{x}, t = r, \\ C_\beta(\mathbf{s}, \mathbf{x}) + C_\kappa^{(s)}(\mathbf{s}, \mathbf{x}), & \text{otherwise.} \end{cases} \quad (6.20)$$

The two spatial covariance functions, C_β and $C_\kappa^{(s)}$, are not identifiable in this model; to deal with this, Stein (1986) assumes that they are individually specified.

The spatio-temporal covariance functions above are representative of models in the early spatio-temporal statistics literature. More recently, there has been considerable development of large classes of spatio-temporal covariance functions; the rest of this section presents an overview.

6.1.1 Theoretical Properties

A covariance function is a *nonnegative-definite function*, and vice versa. Therefore, investigating nonnegative-definite functions in space and time has been an important endeavor in spatio-temporal statistics.

Definition 6.1. A function $\{f(\mathbf{u}, \mathbf{v}) : \mathbf{u}, \mathbf{v} \in D\}$ defined on $D \times D$ is said to be *nonnegative-definite*, if for any complex numbers $\{a_i : i = 1, \dots, m\}$, any $\{\mathbf{u}_i : i = 1, \dots, m\}$ in D , and any integer m , we have

$$\sum_{i=1}^m \sum_{j=1}^m a_i \bar{a}_j f(\mathbf{u}_i, \mathbf{u}_j) \geq 0, \quad (6.21)$$

where recall that $\bar{a}$ denotes the complex conjugate of a .

Positive-definiteness of $f(\cdot, \cdot)$ involves the inequality above being strictly positive whenever $\mathbf{a} \equiv (a_1, \dots, a_m)'$ is a nonzero vector. In this book, we shall be considering, almost exclusively, covariance functions that are positive-definite. We now give some examples of D .

If D is a subset of the integers $\mathbb{Z}$, f is a (temporal) covariance function for a discrete-index time series (Chapter 3), and if $D \subset \mathbb{R}^d$, then f is a (spatial) covariance function for a continuous-index spatial process (Chapter 4). In the spatio-temporal setting, D could be a subset of $\mathbb{R}^d \times \{\dots, -1, 0, 1, \dots\}$ or of $\mathbb{R}^d \times \mathbb{R}$. Hence, a spatial covariance function in $\mathbb{R}^{d+1}$ (see Chapter 4) could serve as a spatio-temporal covariance function in $\mathbb{R}^d \times \mathbb{R}$. In (6.21), we write $\mathbf{u} = (\mathbf{s}; t)$ and $f(\mathbf{u}_i, \mathbf{u}_j) \equiv f((\mathbf{s}_i; t_i), (\mathbf{s}_j; t_j))$, to obtain a spatio-temporal covariance function.

6.1.2 Stationarity in Space or Time

Definition 6.2. We say that f is a *stationary spatio-temporal covariance function* on $\mathbb{R}^d \times \mathbb{R}$, if it satisfies Definition 6.1 and can be written as

$$f((\mathbf{s}; t), (\mathbf{x}; r)) = C(\mathbf{s} - \mathbf{x}; t - r), \quad \mathbf{s}, \mathbf{x} \in \mathbb{R}^d, t, r \in \mathbb{R}.$$

If a random process $Y(\cdot; \cdot)$ has a constant expectation and a stationary covariance function, $C_Y(\mathbf{h}; \tau)$, then it is said to be *second-order (or weakly) stationary*. Strong stationarity of $Y(\cdot; \cdot)$ corresponds to the equivalence of the two probability measures defining the random processes $Y(\cdot; \cdot)$ and $Y(\cdot + \mathbf{h}; \cdot + \tau)$, respectively, for all $\mathbf{h} \in \mathbb{R}^d$ and all $\tau \in \mathbb{R}$.

When there is no ambiguity, we drop dependence on Y and write simply, $C(\mathbf{h}; \tau)$, unless clarity demands otherwise. Upon rewriting (6.21) in terms of $C(\cdot; \cdot)$, we see that the defining property of a stationary spatio-temporal covariance function $C(\cdot; \cdot)$ on $\mathbb{R}^d \times \mathbb{R}$ becomes

$$\sum_{i=1}^m \sum_{j=1}^m a_i \bar{a}_j C(\mathbf{s}_i - \mathbf{s}_j; t_i - t_j) \geq 0, \quad (6.22)$$

for any $\{a_i\}$, any $\{\mathbf{s}_i; t_i\}$, and any m . Notice that $C(\mathbf{h}; 0)$, $\mathbf{h} \in \mathbb{R}^d$, corresponds to a stationary spatial covariance function and that $C(\mathbf{0}; \tau)$, $\tau \in \mathbb{R}$, corresponds to a stationary temporal covariance function.

The *stationary spatio-temporal correlation function* associated with $C(\cdot; \cdot)$ is

$$\rho(\mathbf{h}; \tau) \equiv C(\mathbf{h}; \tau) / C(\mathbf{0}; 0), \quad \mathbf{h} \in \mathbb{R}^d, \tau \in \mathbb{R}, \quad (6.23)$$

an example of which is given by (6.3). The correlation function (6.23) represents spatio-temporal dependence in a continuous-space ($\mathbb{R}^d$), continuous-time ($\mathbb{R}$) model.

The intuitive notion that values tend to be weakly correlated when they are “far apart” can be captured by using covariance functions that tend to zero as their spatio-temporal coordinates move farther apart. Covariance models are often used that are identically zero when separated by more than a fixed distance in space or in time; these are sometimes referred to as *compactly supported* covariances.

It is possible that stationarity of the covariance function be considered separately for space and time. *Spatial stationarity* of the covariance function corresponds to

$$\text{cov}(Y(\mathbf{s}; t), Y(\mathbf{x}; r)) \equiv C(\mathbf{s} - \mathbf{x}; t, r),$$

and *temporal stationarity* of the covariance function corresponds to

$$\text{cov}(Y(\mathbf{s}; t), Y(\mathbf{x}; r)) \equiv C(\mathbf{s}, \mathbf{x}; t - r).$$

Recall that *spatio-temporal stationarity* of the covariance function corresponds to

$$\text{cov}(Y(\mathbf{s}; t), Y(\mathbf{x}; r)) \equiv C(\mathbf{s} - \mathbf{x}; t - r).$$

Now we turn our attention to isotropy. Similar to spatial stationarity, *spatial isotropy* corresponds to

$$\text{cov}(Y(\mathbf{s}; t), Y(\mathbf{x}; r)) \equiv C(\|\mathbf{s} - \mathbf{x}\|; t, r).$$

Geometric spatial anisotropy corresponds to

$$\text{cov}(Y(\mathbf{s}; t), Y(\mathbf{x}; r)) \equiv C(\{p_1(s_1 - x_1)^2 + \cdots + p_d(s_d - x_d)^2\}^{1/2}; t, r),$$

where $p_1 > 0, \dots, p_d > 0$. Finally, *geometric spatio-temporal anisotropy* corresponds to

$$\text{cov}(Y(\mathbf{s}; t), Y(\mathbf{x}; r)) = C(\{p_1(s_1 - x_1)^2 + \cdots + p_d(s_d - x_d)^2 + q(t - r)^2\}^{1/2}),$$

where $p_1 > 0, \dots, p_d > 0, q > 0$.

6.1.3 Separability and Full Symmetry

Definition 6.3. A random process $Y(\cdot; \cdot)$ is said to have a *separable spatio-temporal covariance function* if, for all $\mathbf{s}, \mathbf{x} \in \mathbb{R}^d, t, r \in \mathbb{R}$, we obtain

$$\text{cov}(Y(\mathbf{s}; t), Y(\mathbf{x}; r)) = C^{(s)}(\mathbf{s}, \mathbf{x}) \cdot C^{(t)}(t, r), \quad (6.24)$$

where $C^{(s)}$ and $C^{(t)}$ are spatial and temporal covariance functions, respectively.

As a consequence of (6.24), a simple class of spatio-temporal covariance functions is given by the *product* of individual spatial and temporal covariance functions. When $C^{(s)}$ and $C^{(t)}$ are, respectively, spatially and temporally stationary, (6.24) becomes

$$C(\mathbf{h}; \tau) = C^{(s)}(\mathbf{h}) \cdot C^{(t)}(\tau), \quad \mathbf{h} \in \mathbb{R}^d, \tau \in \mathbb{R}. \quad (6.25)$$

Also, separability implies that the spatio-temporal correlation function $\rho(\cdot; \cdot)$ given by (6.23) satisfies

$$\rho(\mathbf{h}; \tau) = \rho(\mathbf{h}; 0) \cdot \rho(0; \tau), \quad \mathbf{h} \in \mathbb{R}^d, \tau \in \mathbb{R}. \quad (6.26)$$

Clearly the converse is true, and hence (6.26) is a characterization of separability in second-order stationary processes. This is a nice result that allows one to make a visual inspection of whether separability holds, by *comparing* the contours of $C(\mathbf{h}; \tau)$ with those of $C(\mathbf{h}; 0) \cdot C(0; \tau)$. Figure 6.5a is an example of (6.26), where we chose the same parameter values, $\alpha = 1$ and $\beta = 20$, that were used in Figure 6.2. In the figure, $\rho(\mathbf{h}; 0)$ is given by (6.4), and $\rho(0; \tau)$ is given by (6.5). By contrast, Figure 6.5b shows the corresponding spatio-temporal correlation function (6.3) derived from (6.1); this is the same correlation function shown in Figure 6.3. A comparison of the contours in Figure 6.5b with those in Figure 6.5a shows that the diffusion-injection model given by (6.1) does *not* yield a separable covariance function. This is usually the case; spatio-temporal models constructed dynamically (i.e., causally) often have important components of spatio-temporal interaction, for which a separable covariance function is inadequate.

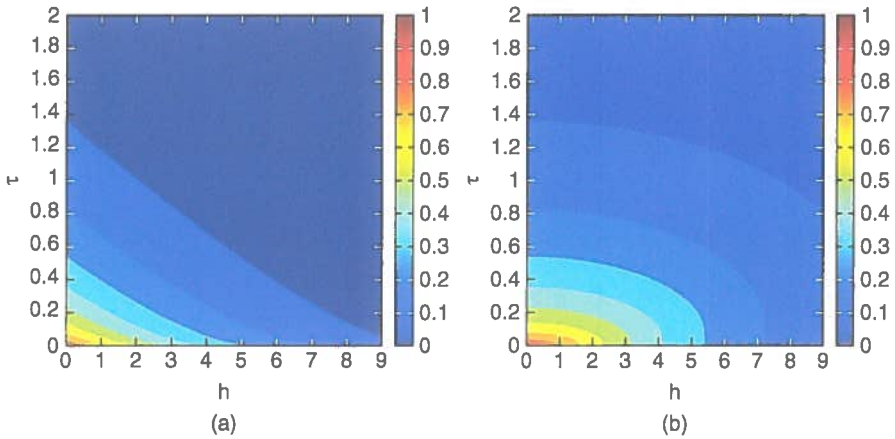


Figure 6.5 (a) Contour plot of separable correlation function, given by (6.26), as a function of spatial lag h and temporal lag τ ; $\alpha = 1$ and $\beta = 20$. (b) Contour plot of nonseparable correlation function (6.3); $\alpha = 1$, $\beta = 20$. (This is for comparison to (a), and it is the same plot as given in Figure 6.3.)

Because of their ease of definition, and because (matrix) computations in spatial statistical analysis can be much simpler, separable covariance models dominated the literature up until the beginning of the twenty-first century. But, they have very particular properties that are rarely seen in empirical studies of spatio-temporal dependence. For example, separability implies that $C(\mathbf{h}_1; \tau) \propto C(\mathbf{h}_2; \tau)$; $\tau \in \mathbb{R}$, which means that each time series has *identical* cross-correlational properties with any other time series at *any* displaced location, both near and far! Likewise, $C(\mathbf{h}; \tau_1) \propto C(\mathbf{h}; \tau_2)$; $\mathbf{h} \in \mathbb{R}^d$. The resulting contour lines for a separable $\{C(\mathbf{h}; \tau): \mathbf{h} \in \mathbb{R}^d, \tau \in \mathbb{R}\}$ show a remarkably regular structure.

To illustrate this, Figure 6.6 shows spatial and temporal profiles of the plots in Figure 6.5. That is, plots of $\rho(\cdot; \tau_i)$ for *separable* $\rho(\cdot; \cdot)$ given by (6.26) and different values of τ_i are shown in Figure 6.6a. Similar plots of $\rho(h_i; \cdot)$ for *separable* $\rho(\cdot; \cdot)$ given by (6.26) and different values of h_i are shown in Figure 6.6b. The corresponding profile plots for *nonseparable* $\rho(\cdot; \cdot)$ given by (6.3) are shown in Figures 6.6c and 6.6d, respectively. Upon comparing Figure 6.6a with Figure 6.6c, and Figure 6.6b with Figure 6.6d, we see the rather regular structure that separability imposes on the profile plots.

Another property of spatio-temporal covariance functions that is sometimes assumed, but which is usually not appropriate, particularly when modeling flow processes, is *full symmetry*.

Definition 6.4. A random process $Y(\cdot; \cdot)$ is said to have a *fully symmetric spatio-temporal covariance function* if, for all $\mathbf{s}, \mathbf{x} \in \mathbb{R}^d$, $t, r \in \mathbb{R}$,

$$\text{cov}(Y(\mathbf{s}; t), Y(\mathbf{x}; r)) = \text{cov}(Y(\mathbf{s}; r), Y(\mathbf{x}; t)). \quad (6.27)$$

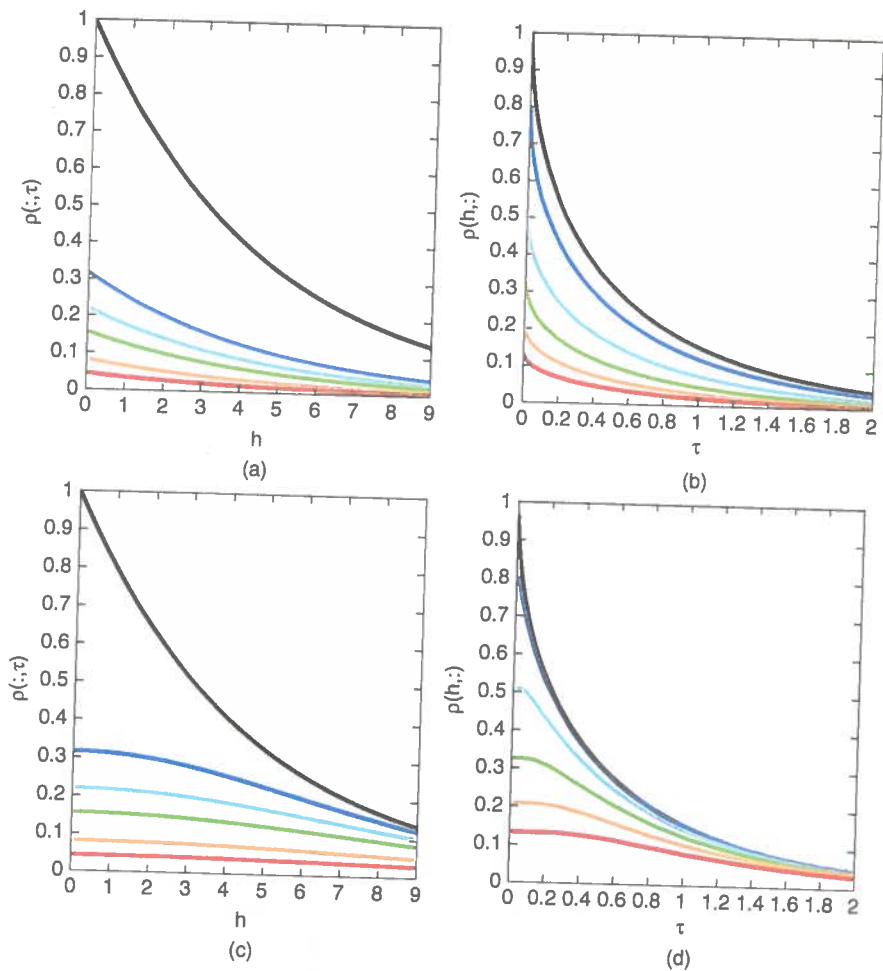


Figure 6.6 Top panels (a) and (b) show a *separable* correlation function: (a) Plot of separable correlation function described in (6.26), versus spatial lag h , for temporal lags $\{\tau_i\} = \{0, 0.5, 0.75, 1.0, 1.5, 2.0\}$; $\alpha = 1$, $\beta = 20$. (b) Plot of separable correlation function described in (6.26), versus temporal lag τ , for spatial lag $\{h_i\} = \{0, 1, 3, 5, 7, 9\}$; $\alpha = 1$, $\beta = 20$. Bottom panels (c) and (d) show a *nonseparable* correlation function: (c) Plot of nonseparable correlation function described in (6.3), versus spatial lag h , for the same temporal lags as in (a); $\alpha = 1$, $\beta = 20$. (d) Plot of nonseparable correlation function described in (6.3), versus temporal lag τ , for the same spatial lags as in (b); $\alpha = 1$, $\beta = 20$.

If Y is a variable like (the average maximum and minimum) daily temperature, (6.27) would say that the covariance between yesterday's temperature in St. Louis, MO, and today's temperature in Columbus, OH, is identical to that between yesterday's temperature in Columbus, and today's temperature in St. Louis. Because St. Louis is west of Columbus, this would be an unlikely spatio-temporal covariance model for daily local temperatures.

For examples of (6.27), we return to the second-order-stationary model given by (6.3). Let $\rho^{(s)}(h); h \geq 0$, be the spatial correlation given by (6.4) and $\rho^{(t)}(\tau); \tau \geq 0$, be the temporal correlation given by (6.5). Then, for $p > 0$,

$$C(h; \tau) \equiv p \cdot \rho^{(s)}(h) \cdot \rho^{(t)}(\tau), \quad (6.28)$$

defines a separable covariance function. Indeed, Figure 6.4b is a contour plot of (6.28) with $p = 1$. Furthermore, for $p > 0$, $q \geq 0$, and $r \geq 0$,

$$C(h; \tau) \equiv p \cdot \rho^{(s)}(h) \cdot \rho^{(t)}(\tau) + q \cdot \rho^{(s)}(h) + r \cdot \rho^{(t)}(\tau) \quad (6.29)$$

is an example of a fully symmetric covariance function, which reduces to a separable covariance function when $q = r = 0$. If $p = 0$, (6.29) yields an additive covariance function, which is nonnegative-definite but not positive-definite. Figure 6.7 gives a contour plot of $\rho(h; \tau) \equiv C(h; \tau)/C(0; 0)$, where $C(\cdot; \cdot)$ is given by (6.29) and hence is fully symmetric; chosen parameters are $p = q = r = 1/3$, and the PDE parameters are $\alpha = 1$ and $\beta = 20$ (same as in Figure 6.2). Further discussion of covariance functions like (6.29), which are sums and products of purely spatial or temporal covariance functions, is given below in Section 6.1.4.

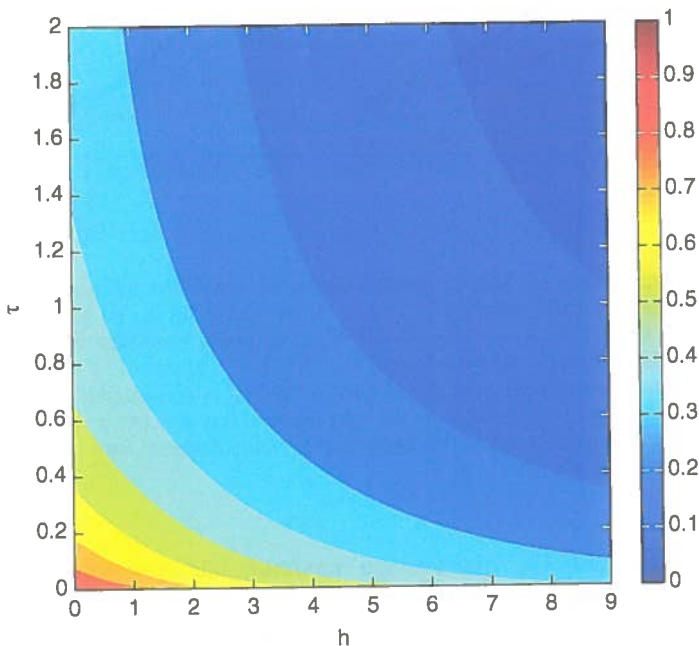


Figure 6.7 Contour plot of fully symmetric correlation function derived from (6.29); $\alpha = 1$, $\beta = 20$, and $p = q = r = \frac{1}{3}$.

Notice that, in general, the class of processes whose covariances are separable, are contained within the class of processes whose covariances are fully symmetric. Figure 6.8 shows several simulations from a mean-zero, stationary, Gaussian, spatio-temporal process with nonseparable covariance function proportional to (6.3); Figure 6.9 shows several simulations corresponding to the fully symmetric covariance function given by (6.29) with $p = q = r = 1/3$; and Figure 6.10 shows several simulations corresponding to the separable covariance function given by (6.28) with $p = 1$. All figures use the same PDE parameters, $\alpha = 1$ and $\beta = 20$ (as in Figure 6.2), the same respective random number seeds, and they are calibrated to correspond to covariances with the same variance, $C(0; 0)$. Notice how difficult it is to distinguish between processes with nonseparable, fully symmetric, and separable covariance functions simply by inspecting their *realizations*. Yet, the difference between the spatio-temporal *covariance functions* shown in Figure 6.3 (nonseparable), Figure 6.7 (fully symmetric), and Figure 6.5a (separable), is apparent.

Because separability implies full symmetry, rejection of the hypothesis of full symmetry implies rejection of the hypothesis of separability. Statistical testing procedures, where the null hypothesis is either separability or is full symmetry, have been considered by Mitchell, Genton, and Gumpertz (2005), Scaccia and Martin (2005), Fuentes (2006), Li, Genton, and Sherman (2007), and Crujeiras, Fernandez-Casal, and González-Manteiga (2010).

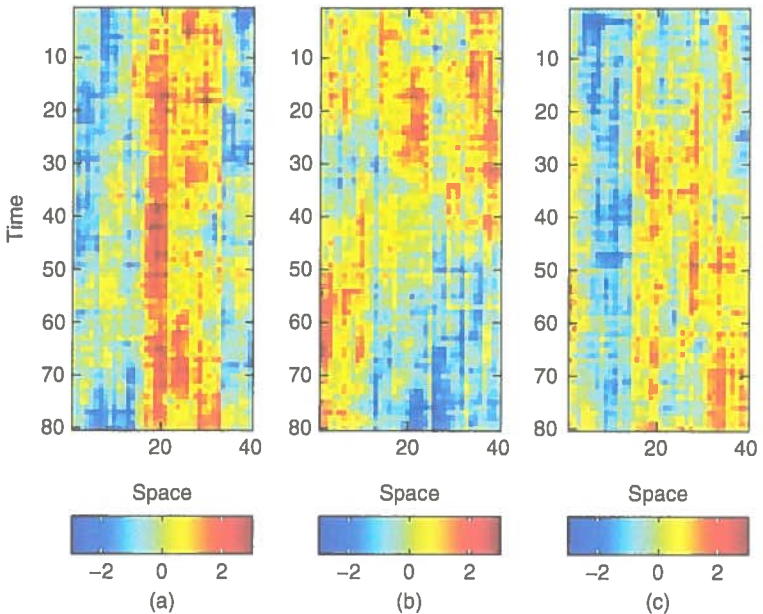


Figure 6.8 The three independent simulations shown in (a), (b), and (c) are from a stationary Gaussian process with mean zero and nonseparable covariance function proportional to (6.3). Here, $s \in D_s \equiv \{\Delta_s, \dots, 40\Delta_s\}$ with $\Delta_s = 1$, and $t \in D_t \equiv \{\Delta_t, \dots, 80\Delta_t\}$ with $\Delta_t = .01$.

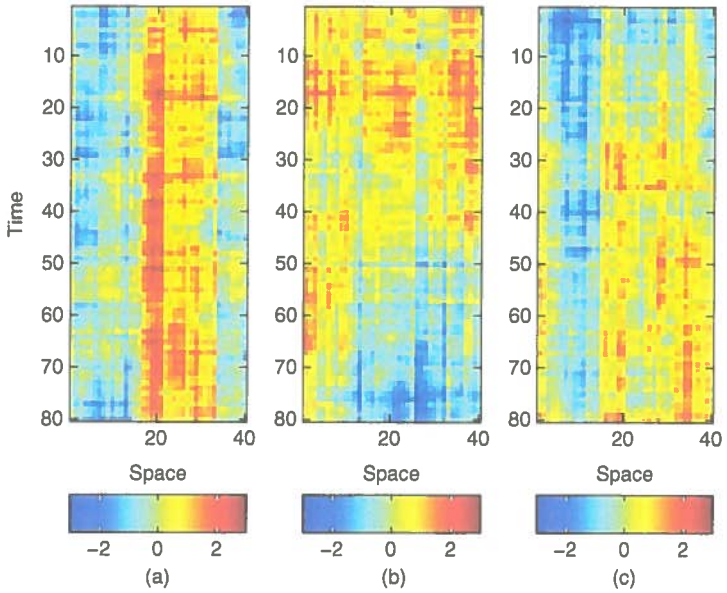


Figure 6.9 The three independent simulations shown in (a), (b), and (c) are from a stationary Gaussian process with mean zero and fully symmetric covariance function given by (6.29); D_s and D_t are the same as in Figure 6.8.

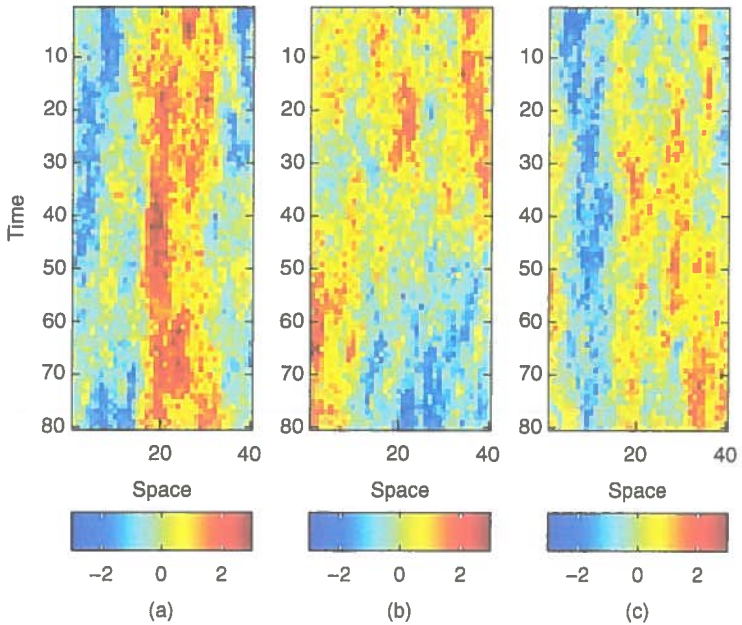


Figure 6.10 The three independent simulations shown in (a), (b), and (c) are from a stationary Gaussian process with mean zero and separable covariance function given by (6.28); D_s and D_t are the same as in Figure 6.8.

6.1.4 Sums and Products of Covariance Functions

Because the sum of two nonnegative-definite functions is nonnegative-definite, the following function is a valid spatio-temporal covariance function:

$$C(\mathbf{h}; \tau) \equiv pC^{(s_1)}(\mathbf{h}) \cdot C^{(t_1)}(\tau) + qC^{(s_2)}(\mathbf{h}) + rC^{(t_2)}(\tau), \quad (6.30)$$

where $p > 0$, $q > 0$, and $r > 0$, $C^{(s_1)}$ and $C^{(s_2)}$ are spatial covariance functions, and $C^{(t_1)}$ and $C^{(t_2)}$ are temporal covariance functions. A special case of this covariance function has already been considered in (6.29). The class of covariance functions given by (6.30) was considered by De Iaco, Myers, and Posa (2001), and it can be derived from a general result given by Gneiting, Genton, and Guttorp (2007):

Theorem 6.1. Let μ be a finite, nonnegative measure on a non-empty set Θ . Suppose that for each $\theta \in \Theta$, $C^{(s_\theta)}$ and $C^{(t_\theta)}$ are stationary covariance functions on $\mathbb{R}^d$ and $\mathbb{R}$, respectively, and suppose that $\int_{\Theta} |C^{(s_\theta)}(\mathbf{0}) \cdot C^{(t_\theta)}(0)| d\mu(\theta) < \infty$. Then,

$$C(\mathbf{h}; \tau) \equiv \int_{\Theta} C^{(s_\theta)}(\mathbf{h}) \cdot C^{(t_\theta)}(\tau) d\mu(\theta), \quad \mathbf{h} \in \mathbb{R}^d, \tau \in \mathbb{R}, \quad (6.31)$$

is a stationary spatio-temporal covariance function.

We have seen in Section 6.1.3 that separability implies full symmetry; Theorem 6.1 shows that the converse is not true. While each component of the mixture on the right-hand side of (6.31) is separable, and hence fully symmetric, $C(\cdot; \cdot)$ given by (6.31) is not separable; however, it is fully symmetric. Finally, (6.30) is a special case of (6.31), since $C^{(s)}(\cdot) = 1$ or $C^{(t)}(\cdot) = 1$ are covariance functions of a random process trivially equal to a single random variable (with unit variance) for all $\mathbf{s} \in D_s$ or all $t \in D_t$, respectively.

6.1.5 Spatio-Temporal Variogram

There is another way to express dependence, related to the covariance function. The *spatio-temporal variogram* of the process $Y(\cdot; \cdot)$ is defined to be

$$\text{var}(Y(\mathbf{s}; t) - Y(\mathbf{x}; r)) \equiv 2\gamma(\mathbf{s}, \mathbf{x}; t, r), \quad (6.32)$$

along with its stationary version, $2\gamma(\mathbf{h}; \tau)$; $\mathbf{h} \in \mathbb{R}^d$, $\tau \in \mathbb{R}$. The quantity γ is called the *semivariogram*. The process $Y(\cdot; \cdot)$ is said to be *intrinsically stationary*, if it has constant expectation and its variogram is stationary. If $Y(\cdot; \cdot)$ is second-order stationary with (stationary) covariance function $C(\mathbf{h}; \tau)$,

then, as in the purely spatial setting (Section 4.1.1), $Y(\cdot; \cdot)$ is intrinsically stationary with semivariogram,

$$\gamma(\mathbf{h}; \tau) = C(\mathbf{0}; 0) - C(\mathbf{h}; \tau), \quad \mathbf{h} \in \mathbb{R}^d, \tau \in \mathbb{R}. \quad (6.33)$$

Similarly to Section 6.1.2, we can define separately the notions of spatial or temporal intrinsic stationarity, and so on, of the variogram. In general, there is no reason why the space dimension and the time dimension should behave alike.

6.1.6 Spectral Representation

Stationary spatio-temporal covariance functions have a spectral representation, in an analogous manner to the purely spatial (Chapter 4) and purely temporal (Chapter 3) stationary covariance functions. In fact, the spectral representation can be viewed simply as Theorem 4.1 (Bochner's Theorem) in $\mathbb{R}^{d+1}$. Written to emphasize space and time separately, it is

$$C(\mathbf{h}; \tau) = \int \int e^{i\mathbf{h}'\boldsymbol{\omega} + i\tau\xi} dF(\boldsymbol{\omega}; \xi), \quad \mathbf{h} \in \mathbb{R}^d, \tau \in \mathbb{R}, \quad (6.34)$$

where F is a finite, nonnegative, symmetric measure on $\mathbb{R}^d \times \mathbb{R}$, called the *spectral measure*. If the spectral measure can be written as $dF(\boldsymbol{\omega}; \xi) = f(\boldsymbol{\omega}; \xi) d\boldsymbol{\omega} d\xi$, then $f(\cdot; \cdot)$ is called the *spectral density*, and $\int f(\boldsymbol{\omega}; \tau) d\boldsymbol{\omega} d\tau = C(\mathbf{0}; 0)$. This happens if C is integrable (i.e., $\int \int |C(\mathbf{h}; \tau)| d\mathbf{h} d\tau < \infty$), and then the inverse result holds:

$$f(\boldsymbol{\omega}; \xi) = (2\pi)^{-(d+1)} \int \int e^{-i\mathbf{h}'\boldsymbol{\omega} - i\tau\xi} C(\mathbf{h}; \tau) d\mathbf{h} d\tau. \quad (6.35)$$

That is, f is the Fourier transform of C (and C is the inverse Fourier transform of f). Furthermore, nonnegativity of f corresponds to nonnegative-definiteness of C , and the two functions are equivalent ways to represent (second-order stationary) spatio-temporal dependence.

The spectral representation (6.34) offers a convenient way to construct valid stationary spatio-temporal covariance functions. The essential property of nonnegative-definiteness is difficult to establish directly from its definition but, through Bochner's Theorem, we only have to establish nonnegativity of $f(\cdot; \cdot)$. A related strategy is to start with a family of nonnegative f and take each member's inverse Fourier transform to obtain a family of nonnegative-definite functions C . In the spatio-temporal context, Cressie and Huang (1999) proved the following theorem:

Theorem 6.2. Suppose that C is a symmetric, continuous, bounded, integrable function on $\mathbb{R}^d \times \mathbb{R}$. Then C is a stationary spatio-temporal covariance

function if and only if,

$$\alpha(\omega; \tau) \equiv (2\pi)^{-d} \int e^{-i\mathbf{h}'\omega} C(\mathbf{h}; \tau) d\mathbf{h}, \quad \tau \in \mathbb{R}, \quad (6.36)$$

is nonnegative-definite in τ , for almost all $\omega \in \mathbb{R}^d$. (Here, “almost all ω ” means that nonnegative-definiteness of (6.36) may fail for some $\omega \in \mathbb{R}^d$, but the Lebesgue measure of the set of all such ω is equal to zero.)

The idea here is that α is only a *one-dimensional* function of τ , for which there are many results for guaranteeing nonnegative-definiteness (e.g., Cressie, 1993, p. 86). One first establishes a suitable family for $\alpha(\omega; \tau)$, where members need only be nonnegative-definite in τ , and $\int |\alpha(\omega; \tau)| d\omega < \infty$. Then, for given τ , the inverse Fourier transform of $\alpha(\cdot; \tau)$,

$$C(\mathbf{h}; \tau) \equiv \int e^{i\mathbf{h}'\omega} \alpha(\omega; \tau) d\omega, \quad \mathbf{h} \in \mathbb{R}^d, \tau \in \mathbb{R}, \quad (6.37)$$

yields a spatio-temporal covariance function. Generally, families constructed in this way will not be separable. (Separability occurs when $\alpha(\omega; \tau) \equiv k(\omega) \cdot \rho(\tau)$.)

Example(Cressie and Huang, 1999): Let

$$\alpha(\omega; \tau) \equiv \exp\{-\|\omega\|^2|\tau|/4\} \exp\{-\delta\tau^2\} \exp\{-c_0\|\omega\|^2/4\},$$

where $\delta > 0$ and $c_0 > 0$. This satisfies the conditions of Theorem 6.2 and results in the following nonseparable family:

$$C(\mathbf{h}; \tau) = \sigma^2 \exp\{-b^2\|\mathbf{h}\|^2/(a^2\tau^2 + 1)\}/(a^2\tau^2 + 1)^{d/2}, \quad \mathbf{h} \in \mathbb{R}^d, \tau \in \mathbb{R}, \quad (6.38)$$

where $\sigma^2 = C(\mathbf{0}; 0)$, and $a \geq 0$ and $b \geq 0$ are scaling parameters for time and space, respectively. Figure 6.11a shows a contour plot of the spatio-temporal covariance function (6.38) in one-dimensional space (i.e., $d = 1$, $a = 2$, $b = 0.2$, and $\sigma^2 = 1$). Figure 6.11b shows a realization from a Gaussian process with mean zero and spatio-temporal covariance function given by (6.38). The smoothness of the realization in space is remarkable, and it is due to the behavior of (6.38) for small $\|\mathbf{h}\|$; notice that it is differentiable in $\|\mathbf{h}\|$ at $\|\mathbf{h}\| = 0$, resulting in smooth sample paths (e.g., Cressie, 1993, p. 60).

Consider the separable spatio-temporal covariance function, $C(\mathbf{h}; 0) \cdot C(\mathbf{0}; \tau)$, where $C(\cdot; \cdot)$ is given by (6.38). (From the discussion in Section 6.1.3, the product is separable and positive-definite.) Figure 6.12a shows a contour plot of this covariance function, and Figure 6.12b shows a realization from the corresponding mean-zero Gaussian process (using the same random number seed as that used in Figure 6.11b). Once again, we see

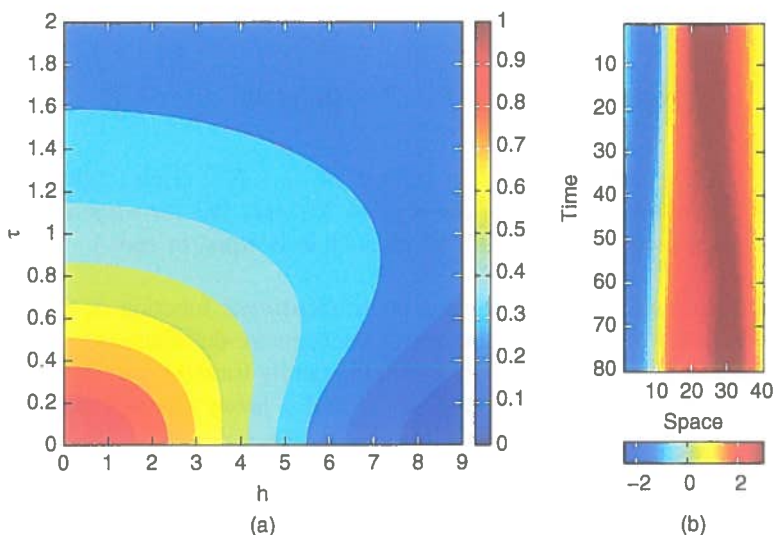


Figure 6.11 (a) Contour plot of the nonseparable covariance function given by (6.38); $d = 1$, $a = 2$, $b = 0.2$, and $\sigma = 1$. (b) Simulation from a stationary Gaussian process with mean zero and the nonseparable covariance function shown in (a). Here, D_s and D_t are the same as in Figure 6.8.

that nonseparability is hard to detect by comparing directly the realizations shown in Figure 6.11b and Figure 6.12b, but their difference is obvious when comparing their covariance functions in Figure 6.11a and Figure 6.12a.

Another construction of $C(\mathbf{h}; \tau)$ based on Theorem 6.2 is due to Gneiting (2002), and his construction does not require integrability of C . He shows that

$$C(\mathbf{h}; \tau) \equiv \sigma^2 \phi(\|\mathbf{h}\|^2 / \psi(|\tau|^2)) / \{\psi(|\tau|^2)\}^{d/2}, \quad \mathbf{h} \in \mathbb{R}^d, \tau \in \mathbb{R}, \quad (6.39)$$

is a valid spatio-temporal covariance function, provided $\{\phi(x) : x \geq 0\}$ is a completely monotone function and $\{\psi(x) : x \geq 0\}$ is a positive function with a completely monotone derivative.

Example(Gneiting, 2002): Choose

$$\begin{aligned} \phi(x) &= \exp\{-px^\gamma\}, & x \geq 0, \quad p > 0, \quad 0 < \gamma \leq 1, \\ \psi(x) &= (qx^\alpha + 1)^\beta, & x \geq 0, \quad q > 0, \quad 0 < \alpha \leq 1, \quad 0 \leq \beta \leq 1. \end{aligned}$$

This results in

$$C(\mathbf{h}; \tau) = \sigma^2 \exp\{-p\|\mathbf{h}\|^{2\gamma} / (q|\tau|^{2\alpha} + 1)^{\beta\gamma}\} / (q|\tau|^{2\alpha} + 1)^{\beta d/2}. \quad (6.40)$$

Suppose we put $\gamma = 1$, $\alpha = 1$, and $\beta = 1$ in (6.40) as a special case; then we obtain the same covariance function given by (6.38) by putting $p = b^2$ and $q = a^2$.

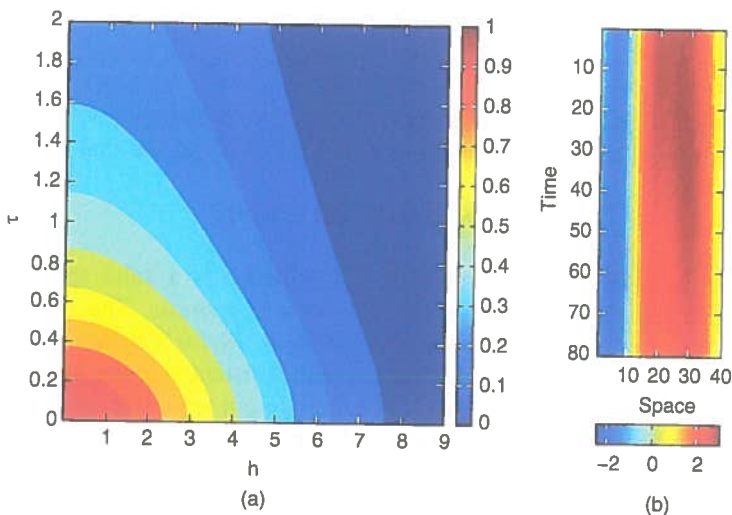


Figure 6.12 (a) Contour plot of the separable covariance function derived from (6.38) as the product of its marginal correlation functions; $d = 1$, $a = 2$, $b = 0.2$, and $\sigma = 1$. (b) Simulation from a stationary Gaussian process with mean zero and the separable covariance function shown in (a). Here, D_s and D_t are the same as in Figure 6.8.

Other examples of $\phi(\cdot)$ and $\psi(\cdot)$ are given by Gneiting (2002) and include, for $x \geq 0$,

$$\begin{aligned}\phi(x) &= (1 + px^\gamma)^{-\nu}; \quad p > 0, \quad 0 < \gamma \leq 1, \quad \nu > 0, \\ \phi(x) &= 2^\nu \{\exp(px^{1/2}) + \exp(-px^{1/2})\}^{-\nu}, \quad p > 0, \quad \nu > 0, \\ \phi(x) &= (2^{\nu-1} \Gamma(\nu))^{-1} (px^{1/2})^\nu K_\nu(px^{1/2}), \quad p > 0, \quad \nu > 0,\end{aligned}$$

where $K_\nu(\cdot)$ is a modified Bessel function of the second kind of order ν (e.g., Abramowitz and Stegun, 1964, pp. 374–379), and

$$\begin{aligned}\psi(x) &= (qx^\alpha + r)/(r(qx^\alpha + 1)), \quad q > 0, \quad 0 < r \leq 1, \quad 0 < \alpha \leq 1, \\ \psi(x) &= \log(qx^\alpha + r)/\log(r), \quad q > 0, \quad r > 1, \quad 0 < \alpha \leq 1.\end{aligned}$$

6.1.7 Taylor's Hypothesis

Sometimes a curious relationship is observed between dependence in space and dependence in time, a relationship that was originally hypothesized for certain processes in fluid dynamics (Taylor, 1938, p. 478).

Definition 6.5. A stationary spatio-temporal covariance function, $C(\mathbf{h}; \tau)$; $\mathbf{h} \in \mathbb{R}^d$, $\tau \in \mathbb{R}$, satisfies *Taylor's hypothesis* if there exists a vector $\mathbf{u}_0 \in \mathbb{R}^d$ such that

$$C(\mathbf{0}; \tau) = C(\mathbf{u}_0 \tau; 0), \quad \tau \in \mathbb{R}, \quad (6.41)$$

where the units of $\mathbf{u}_0$ are (units of $\mathbf{h}$)/(units of τ), and hence $\mathbf{u}_0$ is a velocity vector.

Example: Let $\mathbf{u}_0$ denote a given velocity vector and assume that $C^{(s)}(\cdot)$ is a purely spatial covariance function. Then, it is easy to see that

$$C(\mathbf{h}; \tau) \equiv C^{(s)}(\mathbf{h} - \mathbf{u}_0\tau), \quad (6.42)$$

is nonnegative-definite (Definition 6.1) and hence is a valid spatio-temporal covariance function. It also satisfies Taylor's hypothesis. The process corresponding to (6.42) is sometimes referred to as *Taylor's frozen field*, since it corresponds to a spatial field that moves with constant velocity. Notice that the covariance function given by (6.42) is generally not separable, and it represents another way to construct classes of valid spatio-temporal covariance functions, here based on a purely spatial covariance function.

If $C^o(h)$ is a covariance function in $\mathbb{R}$ such that $C^o(\|\mathbf{h}\|)$ is a valid covariance function in $\mathbb{R}^d$, then the geometrically anisotropic covariance function (Section 6.1.2),

$$C(\mathbf{h}; \tau) \equiv C^o(\{p\|\mathbf{h}\|^2 + q|\tau|^2\}^{1/2}), \quad \mathbf{h} \in \mathbb{R}^d, \tau \in \mathbb{R},$$

where $p > 0$, $q > 0$, is a spatio-temporal covariance function that satisfies Taylor's hypothesis with $\mathbf{u}_0 = (q^{1/2}/p^{1/2}, 0, \dots, 0)'$. Finally, separable models can satisfy Taylor's hypothesis. Again, let $C^o(\cdot)$ be a covariance function in $\mathbb{R}$ such that $C^o(\|\mathbf{h}\|)$ is a valid covariance function in $\mathbb{R}^d$. Then,

$$C(\mathbf{h}; \tau) \equiv C^o(\|\mathbf{h}\|)C^o(|\tau|), \quad h \in \mathbb{R}^d, \tau \in \mathbb{R},$$

satisfies Taylor's hypothesis with any velocity vector $\mathbf{u}_0$ such that $\|\mathbf{u}_0\| = 1$.

6.1.8 Spatio-Temporal Dynamical Models

While the classes of spatio-temporal covariance functions given in this section are quite flexible and can exhibit interaction between space and time, they are derived from moments of bivariate probability distributions, and hence they play a *descriptive* role in representing the spatio-temporal dependence in the process $Y(\cdot; \cdot)$. That is, it is very difficult to look at a covariance function and determine the etiology of the spatio-temporal process under study. We prefer a dynamical specification of the spatio-temporal process, where the dependence is motivated by the evolution of the physical, chemical, biological, economic, and so on, processes through time and in space; see the examples given in Chapter 9. When the model is built mechanistically, any spatio-temporal covariance function derived from the model will automatically be nonnegative-definite. For example, the stochastic diffusion-injection model (6.1) results in a spatio-temporal covariance function given by the stationary

(in space and time) positive-definite function (6.3). This same approach can be taken whether the temporal index t is discrete or continuous; further discussion is given in Section 6.3.1.

6.2 SPATIO-TEMPORAL KRIGING

Apart from characterizing statistical dependencies in space and time, what can spatio-temporal covariance functions be used for? We have already seen in Chapter 4 how optimal linear spatial prediction (kriging) can be formulated based on the processes' spatial statistical dependencies. These are typically quantified through a spatial variogram or a spatial covariance function. In the spatio-temporal context, recall from Section 6.1.5 that the variogram is defined as

$$2\gamma(\mathbf{s}, \mathbf{x}; t, r) \equiv \text{var}(Y(\mathbf{s}; t) - Y(\mathbf{x}; r)), \quad \mathbf{s}, \mathbf{x} \in \mathbb{R}^d, \quad t, r \in \mathbb{R}. \quad (6.43)$$

The goal of kriging is to predict $Y(\mathbf{s}_0; t_0)$ from incomplete and noisy data. Any unbiased linear predictor, $Y^\circledast(\mathbf{s}_0; t_0)$, of $Y(\mathbf{s}_0; t_0)$, has the property that its mean squared prediction error, $E(Y^\circledast(\mathbf{s}_0; t_0) - Y(\mathbf{s}_0; t_0))^2$, can be expressed in terms of the variogram. It can also obviously be expressed in terms of the covariance function, which we shall do below.

Recall the spatio-temporal *covariance function*,

$$C(\mathbf{s}, \mathbf{x}; t, r) \equiv \text{cov}(Y(\mathbf{s}; t), Y(\mathbf{x}; r)), \quad (6.44)$$

and notice that the the semivariogram is easily obtained from the formula,

$$\gamma(\mathbf{s}, \mathbf{x}; t, r) = (1/2)\{C(\mathbf{s}, \mathbf{s}; t, t) + C(\mathbf{x}, \mathbf{x}; r, r)\} - C(\mathbf{s}, \mathbf{x}; t, r).$$

Section 6.1 gives valid (i.e., nonnegative-definite) stationary models for (6.44). Essentially, stationarity is an assumption of convenience for parameter estimation and sometimes for computing; however, it may be too strong an assumption over large spatio-temporal domains.

In fact, kriging does *not* require a stationarity assumption, as we illustrate in the rest of Section 6.2. In what follows, we shall develop spatio-temporal kriging based on the general descriptive specification of dependence given by the covariance function, (6.44). We do it for the continuous-time case, although it is equally appropriate for discrete time. Suppose that the data are

$$Z(\mathbf{s}_i; t_{ij}) = Y(\mathbf{s}_i; t_{ij}) + \varepsilon(\mathbf{s}_i; t_{ij}), \quad j = 1, \dots, T_i, \quad i = 1, \dots, m, \quad (6.45)$$

where $\{\varepsilon(\mathbf{s}_i; t_{ij})\}$ is independent of $Y(\cdot; \cdot)$ and represents the measurement error that is henceforth assumed to be *iid* with mean zero and variance σ_ε^2 . Write $\mathbf{Z}^{(i)} \equiv (Z(\mathbf{s}_i; t_{ij}) : j = 1, \dots, T_i)'; \quad i = 1, \dots, m$. Then we wish to predict the

hidden value $Y(s_0; t_0)$, based on all the data defined by (6.45). This would be appropriate for a retrospective analysis (called a *re-analysis* in the climate sciences). That is, we are not concerned with filtering (Section 8.1) in this discussion of spatio-temporal kriging.

6.2.1 Simple Kriging

The *simple-kriging* (*sk*) predictor, $Y^*(s_0; t_0)$, takes the form of a linear combination of the data:

$$\begin{aligned} Y^*(s_0; t_0) &\equiv \sum_{i=1}^m \sum_{j=1}^{T_i} \ell_{ij} Z(s_i; t_{ij}) + c \\ &\equiv \boldsymbol{\ell}' \mathbf{Z} + c, \end{aligned} \quad (6.46)$$

where $\mathbf{Z} \equiv (\mathbf{Z}^{(1)'} , \dots, \mathbf{Z}^{(m)'})'$, and $\boldsymbol{\ell}$ and c are optimized (i.e., the mean squared prediction error is minimized); see below.

Assume for the moment that the mean function,

$$\mu(s; t) \equiv E(Y(s; t)), \quad s \in D_s, \quad t \in D_t,$$

is *known*. Should the mean function be unknown, different versions of kriging can be developed: If the mean function is assumed constant over space and time, then we can derive the *ordinary-kriging* predictor (Section 6.2.2); if the mean function is a linear combination of covariates, then we can derive the *universal-kriging* predictor (Cressie, 1993, Section 3.4).

From the point of view of kriging, time is simply another dimension, and Section 4.1.2 can be consulted for carrying out kriging in Euclidean spaces. Clearly, any spatio-temporal covariance function (or variogram) would have to respect that distances in time would be treated differently from distances in space. We shall now illustrate this important point with simple kriging, but there are analogous ordinary-kriging and universal-kriging equations that are straightforward to derive in the spatio-temporal setting (Section 6.2.2). Write $\mathbf{C}_Z \equiv \text{var}(\mathbf{Z})$, $\mathbf{c}'_0 \equiv \text{cov}(Y(s_0; t_0), \mathbf{Z})$, and $C_{0,0} \equiv \text{var}(Y(s_0; t_0))$. If both the hidden process $Y(\cdot; \cdot)$ and the measurement-error process $\varepsilon(\cdot; \cdot)$ are assumed to be Gaussian processes and we write $\mathbf{Z} = \mathbf{Y} + \boldsymbol{\varepsilon}$, then $\mathbf{Y}$ and $\boldsymbol{\varepsilon}$ follow Gaussian distributions with covariance matrices, $\boldsymbol{\Sigma}_Y$ and $\boldsymbol{\Sigma}_\varepsilon$, respectively. From (6.45) and Section 4.1.2, we have

$$\begin{bmatrix} Y(s_0; t_0) \\ \mathbf{Z} \end{bmatrix} \sim \text{Gau} \left(\begin{bmatrix} \mu(s_0; t_0) \\ \boldsymbol{\mu} \end{bmatrix}, \begin{bmatrix} C_{0,0} & \mathbf{c}'_0 \\ \mathbf{c}_0 & \mathbf{C}_Z \end{bmatrix} \right), \quad (6.47)$$

where $\boldsymbol{\mu} \equiv (\mu(s_i; t_{ij}) : j = 1, \dots, T_i, i = 1, \dots, m)'$, and $\mathbf{C}_Z = \boldsymbol{\Sigma}_Y + \boldsymbol{\Sigma}_\varepsilon$. Thus, the posterior distribution is (e.g., Rencher, 2002, p. 88),

$$Y(s_0; t_0) | \mathbf{Z} \sim \text{Gau}(\mu(s_0; t_0) + \mathbf{c}'_0 \mathbf{C}_Z^{-1} (\mathbf{Z} - \boldsymbol{\mu}), C_{0,0} - \mathbf{c}'_0 \mathbf{C}_Z^{-1} \mathbf{c}_0). \quad (6.48)$$

Recall that the *simple-kriging predictor* of $Y(s_0; t_0)$ takes the form $Y^*(s_0; t_0) = \ell'Z + c$, where ℓ and c are chosen to minimize the mean squared prediction error, $E(Y(s_0; t_0) - \ell'Z - c)^2$. Under the joint-Gaussian assumption (6.47), it is easy to see that the simple-kriging predictor is just the posterior mean of (6.48). That is,

$$Y^*(s_0; t_0) = E(Y(s_0; t_0)|Z) = \mu(s_0; t_0) + c_0' C_Z^{-1} (Z - \mu). \quad (6.49)$$

The simple kriging variance is the minimized mean squared prediction error; under the joint-Gaussian assumption (6.47), this is just the posterior variance. That is,

$$\sigma_{sk}^2(s_0; t_0) \equiv E(Y(s_0; t_0) - Y^*(s_0; t_0))^2 = C_{0,0} - c_0' C_Z^{-1} c_0. \quad (6.50)$$

Now drop the joint Gaussian assumption; it is easy to see that (6.49) and (6.50) are still appropriate formulas for the simple-kriging predictor and its mean squared prediction error (Section 4.1.2).

Notice that the kriging equations involve taking the inverse of the $T_+ \times T_+$ positive-definite matrix $C_Z = \text{var}(Z)$, where $T_+ \equiv \sum_{i=1}^m T_i$, but this is often impossible for the sizes of spatio-temporal datasets one sees in geophysical and environmental applications. If C_Z could be simplified, then the impossible might become possible. This practical limitation to spatio-temporal kriging is one reason why separable covariance functions (Section 6.1.3) have been chosen to represent covariances between the data; then C_Z has block structure that makes computation of C_Z^{-1} much easier (see Section 6.2.2). However, separability is not necessary for fast computations; Cressie, Shi, and Kang (2010) use a spatio-temporal random effects model that is nonseparable (and nonstationary), for which kriging is possible, even for massive spatio-temporal datasets.

In the spatio-temporal setting, there are two different ways of writing the data vector Z . In (6.46), we have written it as a multivariate spatial process, where $Z^{(i)}$ is the T_i -dimensional vector of temporal data at the i th site; $i = 1, \dots, m$. Alternatively, we could collect all the data at time point t and call it Z_t . The concatenation of such $\{Z_t : t = 1, \dots, T\}$ amounts to a reordering of the data vector Z in (6.46).

For illustrative purposes, suppose that one datum is observed at each location $\{s_i : i = 1, \dots, m\}$ and at each time $t = 1, \dots, T$. Then writing $Z = (Z^{(1)'}, \dots, Z^{(m)'})'$ is motivated by a model that assumes a *multivariate* (in time) *spatial* process. There is a large literature on multivariate spatial prediction and cokriging (e.g., Myers, 1982; Ver Hoef and Cressie, 1993; Wackernagel, 1995). The optimal prediction itself is quite straightforward. Since cokriging takes an EHM approach, there remains the problem of building and fitting valid cross-variograms, for substitution into the cokriging equations. Section 4.1.5 discusses how difficult it can be to find flexible classes of valid multivariate spatial models, except when the cross-dependence is modeled conditionally.

The data could also be written as $(Z_1', \dots, Z_T')'$, which is motivated by a model that assumes a *multivariate* (in space) *time series* (e.g., the VAR(1)

model presented in Section 3.4.7). Which of the two orderings of data in $\mathbf{Z}$ to use will generally be determined by how the physical process is believed to behave, and by the parsimony of parameterization that is achievable.

6.2.2 Ordinary Kriging

We now consider kriging for the case where $Y(\cdot; \cdot)$ has *constant* unknown mean μ ; this is called *ordinary kriging* (*ok*). Then $\boldsymbol{\mu} = \mu \mathbf{1}$, where $\mathbf{1}$ is a T_+ -dimensional vector of ones. From Section 4.1.2, substituting the generalized-least-squares estimator,

$$\hat{\boldsymbol{\mu}}_{gls} \equiv (\mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{1})^{-1} \mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{Z}, \quad (6.51)$$

for $\mu(\cdot; \cdot) \equiv \mu$ into (6.49) yields the ordinary-kriging predictor,

$$\begin{aligned} \hat{Y}(\mathbf{s}_0; t_0) &\equiv \hat{\boldsymbol{\mu}}_{gls} + \mathbf{c}_0' \mathbf{C}_Z^{-1} (\mathbf{Z} - \hat{\boldsymbol{\mu}}_{gls} \mathbf{1}) \\ &\equiv \boldsymbol{\lambda}' \mathbf{Z}, \end{aligned} \quad (6.52)$$

where $\boldsymbol{\lambda}' \equiv \{\mathbf{c}_0 + \mathbf{1}(1 - \mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{c}_0) / (\mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{1})\}' \mathbf{C}_Z^{-1}$. Formula (6.52) gives the predictor that minimizes the mean squared prediction error over all linear unbiased predictors. Clearly, it takes the same form as ordinary kriging in the purely spatial setting; see (4.51). The minimized mean squared prediction error is the ordinary kriging variance,

$$\sigma_{ok}^2(\mathbf{s}_0; t_0) \equiv C_{0,0} - \mathbf{c}_0' \mathbf{C}_Z^{-1} \mathbf{c}_0 + (1 - \mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{c}_0)^2 / (\mathbf{1}' \mathbf{C}_Z^{-1} \mathbf{1}). \quad (6.53)$$

There are also versions of (6.52) and (6.53) that could be derived for universal kriging (*uk*) in the spatio-temporal setting, but we omit them here.

We apply ordinary kriging to several spatio-temporal datasets obtained by subsampling the realization shown in Figure 6.9a, which recall was generated from a mean-zero Gaussian process with covariance function (6.3). Figures 6.13, 6.14, and 6.15 show the effect of different sampling locations and different sample sizes. Figure 6.13a shows the true surface, $\{Y(\mathbf{s}_0; t_0): \mathbf{s}_0 \in D_s, t_0 \in D_t\}$, and Figure 6.13b shows the kriged surface, $\{\hat{Y}(\mathbf{s}_0; t_0): \mathbf{s}_0 \in D_s, t_0 \in D_t\}$, based on a subsampling of the true surface. Notice that we use the true spatio-temporal covariance function (6.3) when calculating $\hat{Y}(\mathbf{s}_0; t)$ in (6.52). Since there is no measurement error in this example, $\mathbf{Z} = \mathbf{Y}$, and $\hat{Y}(\mathbf{s}_i; t_{ij}) \equiv Z(\mathbf{s}_i; t_{ij})$ is an exact interpolator through the data. Comparison of Figure 6.13b (kriged surface) with Figure 6.13a (true surface) shows how much smoother the kriged surface is. Figure 6.13c shows the kriging-standard-error surface $\{\sigma_{ok}(\mathbf{s}_0; t_0): \mathbf{s}_0 \in D_s, t_0 \in D_t\}$ obtained from (6.53); the farther $(\mathbf{s}_0; t_0)$ is from data locations, the larger the kriging standard error. Figures 6.14 and 6.15 are similarly interpreted.

Notice the ubiquitousness of $\mathbf{C}_Z^{-1}$ in all the kriging formulas. Pragmatic assumptions are sometimes made to compute the inverse of $\mathbf{C}_Z$. For example,

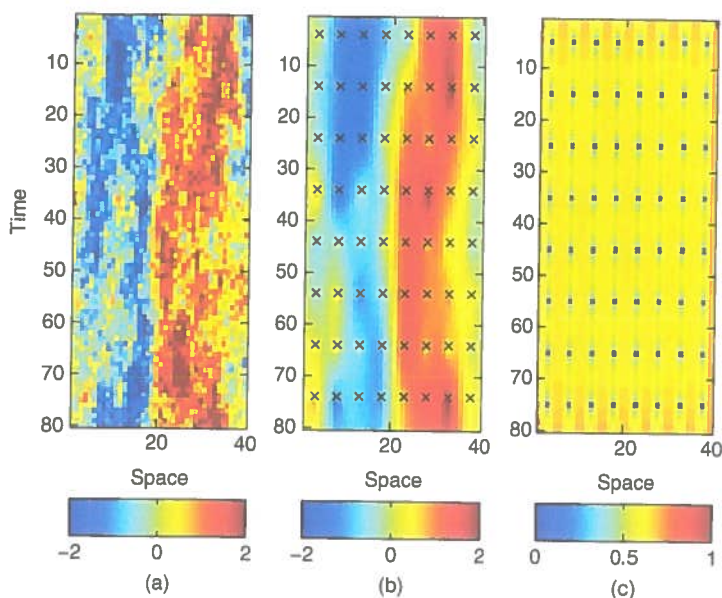


Figure 6.13 (a) Realization from a stationary Gaussian process with mean zero and covariance function derived from (6.3); $\alpha = 1$ and $\beta = 20$. (b) Kriged surface based on $n = 64$ observations; the black crosses denote the locations of the $n = 64$ regularly spaced observations subsampled from the realization in (a). (c) Kriging-standard-error surface; smaller standard errors occur near the observation locations. Here, D_s and D_t are the same as in Figure 6.8.

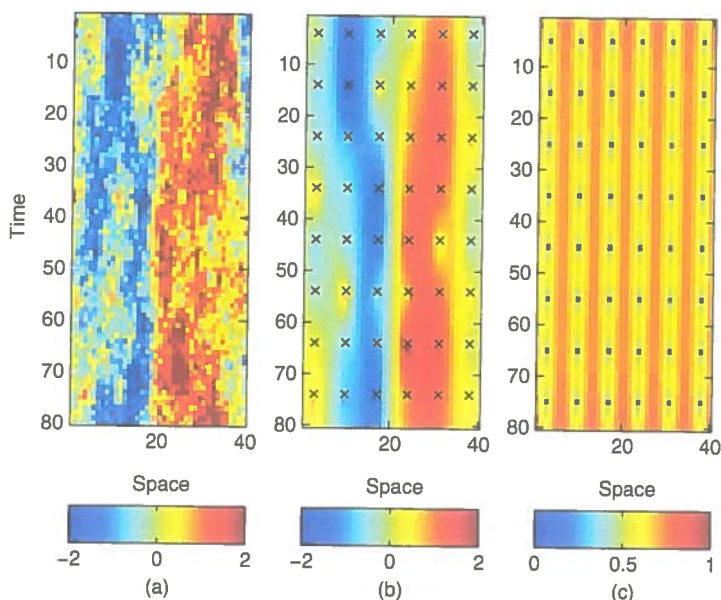


Figure 6.14 (a) Same realization as given in Figure 6.13a. (b) Kriged surface based on $n = 48$ observations; the black crosses denote the locations of the $n = 48$ regularly spaced observations subsampled from the realization in (a). (c) Kriging-standard-error surface; smaller standard errors occur near the observation locations. Here, D_s and D_t are the same as in Figure 6.8.

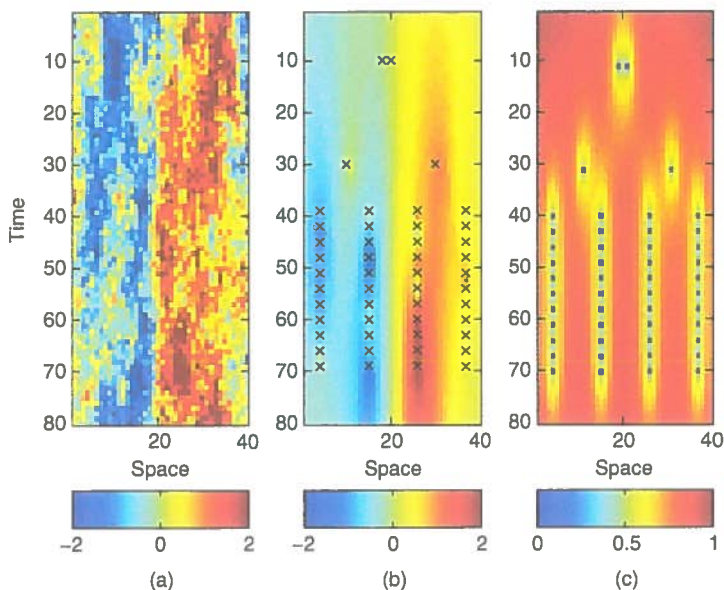


Figure 6.15 (a) Same realization as given in Figure 6.13a. (b) Kriged surface based on $n = 48$ observations; the black crosses denote the locations of the $n = 48$ irregularly spaced observations subsampled from the realization in (a). (c) Kriging-standard-error surface; smaller standard errors occur near the observation locations. Here, D_s and D_t are the same as in Figure 6.8.

consider a spatio-temporal dataset where a datum, $Z(\mathbf{s}_i; t)$, is observed at each location $\mathbf{s}_i$ and at each time point t ; $i = 1, \dots, m$, $t = 1, \dots, T$. One pragmatic assumption is that space and time are separable when modeling the covariation of Z ; that is,

$$\mathbf{C}_Z = \mathbf{C}_Z^{(s)} \otimes \mathbf{C}_Z^{(t)}, \quad (6.54)$$

where $\otimes$ is the Kronecker product, $\mathbf{C}_Z^{(s)}$ is an $m \times m$ covariance matrix of purely spatial covariances, $\mathbf{C}_Z^{(t)}$ is a $T \times T$ covariance matrix of purely temporal covariances, and $T_+ = mT$. In this special case, where $\mathbf{C}_Z$ is given by (6.54), we have

$$\mathbf{C}_Z^{-1} = (\mathbf{C}_Z^{(s)})^{-1} \otimes (\mathbf{C}_Z^{(t)})^{-1}, \quad (6.55)$$

and hence the matrix inverses in (6.49) through (6.53) involve inverses of $m \times m$ and $T \times T$ matrices, which are of much smaller order than the $(mT) \times (mT)$ matrix $\mathbf{C}_Z$. Generally, in the spatio-temporal (and spatial) setting, inverting $\mathbf{C}_Z$ quickly becomes problematic without some form of separable model, or dimension reduction, or approximation of $\mathbf{C}_Z^{-1}$ [Cressie, Shi, and Kang (2010, Section 1) give a discussion of these three approaches].

6.2.3 Optimal Linear Prediction (Kriging) for Dynamical Models

From the kriging equations in this section, it is clear that once the spatio-temporal covariance entries are obtained from the chosen covariance model (see Section 6.1 for examples of such models), kriging does not take notice of the spatio-temporal setting. That is, we might as well be predicting in $\mathbb{R}^{d+1}$, instead of in $\mathbb{R}^d(\text{space}) \times \mathbb{R}(\text{time})$. In contrast, spatio-temporal prediction (smoothing, filtering, forecasting) from a dynamical model exploits the sequential nature of the time component. Broadly speaking, a dynamical approach can finesse the need to take the inverse of the $mT \times mT$ covariance matrix Σ_Z . For example, the Kalman filter computes $Y^*(s_0; t_0)$ and $\sigma_{sk}(s_0; t_0)$ from data $\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_{t_0}$, sequentially. At any time step in the sequence, only matrices of order $m \times m$ need to be inverted. This too may be problematic if m is large. Extensive details are given in Section 8.2, and Section 8.4 presents optimal *nonlinear* prediction in the context of dynamical hierarchical statistical models.

6.3 STOCHASTIC DIFFERENTIAL AND DIFFERENCE EQUATIONS

One of the themes of this book is to use scientific, dynamical relationships where possible, when analyzing spatio-temporal data. Incorporating uncertainty into those relationships is another theme, and we saw in the introduction to this chapter how deterministic PDEs can be made stochastic. Processes that solve stochastic PDEs are (by definition) stochastic, and in this section we discuss some of their statistical properties. We also consider difference equations that are discrete-time approximations to the continuous-time PDEs, and we discuss some of their statistical properties. This section is meant to be motivation for a much more complete discussion that can be found in Sections 7.2 and 7.3.

6.3.1 Spatio-Temporal Statistical Properties

A spatio-temporal stochastic process, $\{Y(\mathbf{s}; t) : \mathbf{s} \in D_s, t \in D_t\}$ (continuous time) or $\{Y_t(\mathbf{s}) : \mathbf{s} \in D_s, t = 0, 1, \dots\}$ (discrete time), is characterized by its finite-dimensional distributions, which are the set of all joint distributions of Y evaluated at any finite subset of $D_s \times D_t$. In practice, it is not easy to derive these joint distributions without further assumptions on the stochastic process. For example, if the process Y is assumed Gaussian, then it is enough to derive just the first two moments of the stochastic process, namely the mean function and the covariance function.

Conversely, if a mean function and a covariance function are specified (Section 6.1), and Y is assumed to be a Gaussian stochastic process with those mean and covariance functions, then obviously any finite-dimensional distribution's mean vector and covariance matrix are fully specified, and that distribution is multivariate Gaussian. This is an example of a valid statistical model that can be used to *describe* spatio-temporal variability.

However, our preference is to *explain* the etiology of the spatio-temporal variability, and we attempt to do this by establishing “physical statistical” relationships. For example, the diffusion-injection model given by (6.1) is a stochastic PDE that could be used to describe the concentration, $Y(s; t)$, of some nutrient in soil at location s and time t . The second derivative with respect to the spatial coordinate represents physical diffusion of the nutrient, which tends to equalize concentration, whereas the term $-\alpha Y(s; t)$ serves to remove nutrient due to oxidation, leaching, or extraction by plants. The “injection” part of the model is given by $\delta(s; t)$, which represents small random additions/removals of the nutrient (Whittle, 1986, p. 431). We shall revisit this model and its statistical properties in Section 6.3.2. Other examples can be found in Ecology, where PDEs are often used; Wikle (2003b) investigates their discrete-time, discrete-space approximations. In this subsection, a more general discussion of the statistical properties of stochastic PDEs and their discrete-time approximations is given.

Here we follow Brown et al. (2000), who give an excellent presentation of how continuous-time models can be limiting cases of discrete-time models. Let $W(\cdot; t)$ denote a continuous-time Brownian motion of spatial processes. That is, W has independent increments in time, and, for any given t , $dW(\cdot; t)$ is a Gaussian spatial process with mean zero and covariance function that we write as $C_\delta(s, \mathbf{x}) dt$; $s, \mathbf{x} \in D_s \subset \mathbb{R}^d$.

Our intention is to use $W(\cdot; t)$ to describe the noise process in a *stochastic* PDE; formally, we write $dW(\cdot; t) = \delta(\cdot; t) dt$. Then $C_\delta(s, \mathbf{x})$ is the *spatial* covariance function of $\delta(\cdot; \cdot)$. For example, $C_\delta(s, \mathbf{x}) = \sigma_\delta^2 I(s = \mathbf{x})$, is a model of spatial white noise that is often used in stochastic PDEs such as (6.1); then δ has the characteristics of white noise in *both* time and space.

Let $\{X(s) : s \in D_s\}$ denote a spatial process in $D_s \subset \mathbb{R}^d$. Define

$$\begin{aligned} \frac{\partial X(s)}{\partial s} &\equiv \left(\frac{\partial X(s)}{\partial s_1}, \dots, \frac{\partial X(s)}{\partial s_d} \right)', \quad \text{and} \\ \frac{\partial^2 X(s)}{\partial s \partial s'} &\equiv \left(\frac{\partial X(s)}{\partial s_i \partial s_j} : i = 1, \dots, d; j = 1, \dots, d \right), \end{aligned}$$

which are, respectively, the d -dimensional vector process of first derivatives and the $d \times d$ matrix process of Hessians. Brown et al. (2000) consider the following stochastic PDE:

$$\frac{\partial Y(s; t)}{\partial t} = (-1/2)\{A Y(\cdot; t)\}(s) + \delta(s; t), \quad (6.56)$$

where $\delta(\cdot; \cdot)$ is a Gaussian stochastic process that is temporally “white” (i.e., uncorrelated) and spatially “colored” (i.e., correlated), with covariance function

C_δ ; and A is a spatial operator defined as follows:

$$\{A X(\cdot)\}(\mathbf{s}) = 2\mu' \frac{\partial X(\mathbf{s})}{\partial \mathbf{s}} - \text{tr} \left\{ \frac{\partial^2 X(\mathbf{s})}{\partial \mathbf{s} \partial \mathbf{s}'} \right\} \Sigma + 2\lambda X(\mathbf{s}),$$

depending on parameters μ (translation rate), Σ ("blurring" rate), and λ (decay rate).

The special case of $\mu = \mathbf{0}$, $\Sigma = 2\beta\mathbf{I}$, $\lambda = \alpha$, and $C_\delta(\mathbf{s}, \mathbf{x}) = \sigma_\delta^2 I(\mathbf{s} = \mathbf{x})$ gives

$$\frac{\partial Y(\mathbf{s}; t)}{\partial t} = \beta \sum_{i=1}^d \frac{\partial^2 Y(\mathbf{s})}{\partial s_i^2} - \alpha Y(\mathbf{s}; t) + \delta(\mathbf{s}; t),$$

which is the d -dimensional version of the diffusion-injection equation (6.1). Other linear stochastic PDEs are given in the introduction to this chapter and in Section 7.2; some nonlinear stochastic PDEs are discussed in Section 7.3.

Brown et al. (2000) show that $Y(\mathbf{s}; t)$ given by (6.56) can be written as

$$Y(\mathbf{s}; t) = \int_0^\infty \exp(-\lambda v) \phi(\mathbf{s}; v\mu, v\Sigma) \circ \delta(\mathbf{s}; t - v) dv, \quad (6.57)$$

where " $\circ$ " denotes the spatial convolution operator,

$$f_1(\mathbf{s}) \circ f_2(\mathbf{s}) \equiv \int f_1(\mathbf{x}) f_2(\mathbf{s} - \mathbf{x}) d\mathbf{x};$$

and $\phi(\cdot; v\mu, v\Sigma)$ is a d -variate Gaussian density with mean $v\mu$ and covariance matrix $v\Sigma$. They also show that the function

$$G(\mathbf{s}; t) \equiv \exp(-\lambda t) \phi(\mathbf{s}; t\mu, t\Sigma) I(0 \leq t < \infty)$$

is the Green's function (or point-spread function) for $Y(\cdot; \cdot)$. Consequently, the covariance function C_Y of Y can be obtained from the covariance function C_δ of the error process δ , as follows.

Suppose that δ is spatially (second-order) stationary (Section 4.1.1), with stationary covariance function $C_\delta(\mathbf{h})$. Then the covariance function of Y is stationary in space and time, and

$$C_Y(\mathbf{h}; \tau) = \int_0^\infty \exp\{-\lambda(2v + |\tau|)\} \phi(\mathbf{h}; \tau\mu, (2v + |\tau|)\Sigma) \circ C_\delta(\mathbf{h}) dv. \quad (6.58)$$

In general, $C_Y(\mathbf{h}; t) \neq C_Y(\mathbf{h}; 0)C_Y(\mathbf{0}; \tau)/C_Y(\mathbf{0}; 0)$, reinforcing our earlier statement that separability is not a common occurrence in physically based models. Furthermore, the spectral density of Y is

$$f_Y(\omega; \xi) = |f_G(\omega; \xi)|^2 f_\delta(\omega), \quad (6.59)$$

where $f_\delta(\omega)$ is the spectral density obtained from $C_\delta(\mathbf{h})$ using Bochner's Theorem (Section 4.1.1), and $f_G(\omega; \xi)$ is the Fourier transform of $G(\mathbf{h}; \tau)$.

In conclusion, Brown et al. (2000) show that if Y satisfies the linear stochastic PDE (6.56), then it can be written as (6.57). Since the error process δ in (6.56) is Gaussian, so too is Y . And if δ is spatially stationary, then Y has covariance function given by (6.58) and spectral density given by (6.59). Brown et al. (2000) derive other expressions for Y and C_Y in terms of the operator A , which we do not present here.

Discrete-time processes can arise from continuous-time processes, due to sampling or aggregation from continuous-time processes, or simply because the process is defined that way (e.g., tree-ring width is a process indexed by a discrete time unit of one year). If, for example, the stochastic PDE (6.56) was discretized in time, then we would obtain the approximation,

$$Y(\mathbf{s}; t) \simeq \{(I - (\Delta_t/2)A_{\Delta_t})Y(\cdot; t - \Delta_t)\}(\mathbf{s}) + \delta(\mathbf{s}; t), \quad (6.60)$$

where here I is the identity operator and A_{Δ_t} is a linear operator defined by

$$\{(I - (\Delta_t/2)A_{\Delta_t})X(\cdot)\}(\mathbf{s}) = \exp(-\lambda\Delta_t) \int \phi(\mathbf{x}; \Delta_t, \boldsymbol{\mu}, \Delta_t, \boldsymbol{\Sigma}) X(\mathbf{s} - \mathbf{x}) d\mathbf{x}.$$

In the expression above, if the spatial domain is discretized to become $D_s = \{\mathbf{s}_1, \dots, \mathbf{s}_n\}$, and terms involving boundary effects and discretization error are dropped, then from (6.60) we obtain the multivariate, discrete-time (Markov) process,

$$\mathbf{Y}_t \simeq \mathbf{M}\mathbf{Y}_{t-\Delta_t} + \boldsymbol{\delta}_t, \quad t = \Delta_t, 2\Delta_t, \dots, \quad (6.61)$$

where $\mathbf{Y}_t \equiv (Y(\mathbf{s}_1; t), \dots, Y(\mathbf{s}_n; t))'$ is the n -dimensional process vector, $\mathbf{Y}_{t-\Delta_t}$ is independent of $\boldsymbol{\delta}_t$, $\boldsymbol{\delta}_t \sim iid \text{Gau}(\cdot, \cdot)$, and $\mathbf{M}$ is an $n \times n$ matrix whose entries can be identified from (6.60). In Section 6.3.2 below, we identify $\mathbf{M}$ for the special case of $\boldsymbol{\mu} = \mathbf{0}$, $\boldsymbol{\Sigma} = 2\beta\mathbf{I}$, and $\lambda = \alpha$ in one-dimensional space, and we show that spatio-temporal covariances derived from (6.61) approximate the covariance function (6.58) very well. Other linear and nonlinear stochastic PDEs are discretized in Sections 7.2.4 and 7.3, respectively.

Consider a discrete-time process $\{Y_t(\mathbf{s}): \mathbf{s} \in D_s, t = 0, 1, \dots\}$. During the time interval from $t-1$ to t , $Y_{t-1}(\mathbf{x})$ can influence the behavior of $Y_t(\mathbf{s})$, where $\mathbf{x}$ could come from near (local weather) or far (teleconnection). The integration of all these influences over $\mathbf{x}$ leads to an integro-difference equation (IDE):

$$Y_t(\mathbf{s}) = \int_{D_s} m(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta}) Y_{t-1}(\mathbf{x}) d\mathbf{x},$$

where $m(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta})$ represents the “influence” of $Y_{t-1}(\mathbf{x})$ on $Y_t(\mathbf{s})$; see Section 7.2.5, where m is called a redistribution kernel that depends on parameters $\boldsymbol{\theta}$. To make the IDE stochastic, we write

$$Y_t(\mathbf{s}) = \int_{D_s} m(\mathbf{s}, \mathbf{x}; \boldsymbol{\theta}) Y_{t-1}(\mathbf{x}) d\mathbf{x} + \eta_t(\mathbf{s}), \quad \mathbf{s} \in D_s, \quad (6.62)$$

where, like δ in (6.56), $\{\eta_t(\mathbf{s}) : \mathbf{s} \in D_s, t = 1, 2, \dots\}$ is uncorrelated in time but correlated in space, and it is often assumed to be a stationary Gaussian process.

Here we follow Storvik, Frigessi, and Hirst (2002), who give results that determine when classes of second-order spatio-temporal covariance functions can be represented according to (6.62) with $D_s = \mathbb{R}^d$ and stationary redistribution kernel, $m(\mathbf{x} - \mathbf{s})$, where for convenience dependence on $\boldsymbol{\theta}$ is suppressed. Then (6.62) can be written as

$$Y_t(\mathbf{s}) = \int m(\mathbf{v}) Y_{t-1}(\mathbf{s} + \mathbf{v}) d\mathbf{v} + \eta_t(\mathbf{s}), \quad \mathbf{s} \in D_s. \quad (6.63)$$

Let $C_\eta(\mathbf{h})$ denote the second-order stationary spatial covariance function of $\eta_t(\cdot)$, and denote its spectral density as $f_\eta(\boldsymbol{\omega})$. Then from (6.63), Y is second-order stationary, and we denote its covariance function as $C_Y(\mathbf{h}; \tau)$. Hold τ fixed and let $f_{Y,\tau}(\boldsymbol{\omega})$ denote the Fourier transform of $C_Y(\mathbf{h}; \tau)$; $\tau = 0, \pm 1, \pm 2, \dots$. Finally, let $f_m(\boldsymbol{\omega})$ be the Fourier transform of the kernel $m(\mathbf{h})$. Then Storvik, Frigessi, and Hirst (2002) show that, for $\boldsymbol{\omega} \in \mathbb{R}^d$,

$$f_{Y,0}(\boldsymbol{\omega}) = \frac{f_\eta(\boldsymbol{\omega})}{1 - f_m(\boldsymbol{\omega})f_m(-\boldsymbol{\omega})}$$

and

$$f_{Y,\tau}(\boldsymbol{\omega}) = f_{Y,0}(\boldsymbol{\omega})\{f_m(-\boldsymbol{\omega})\}^{|\tau|}, \quad \tau = 0, \pm 1, \pm 2, \dots \quad (6.64)$$

Consequently, $C_Y(\mathbf{h}; \tau)$ can be obtained by computing (numerically or analytically) the inverse Fourier transform of the right-hand side of (6.64). In some cases, an analytical result is possible; Storvik, Frigessi, and Hirst (2002) give an example of a spatio-temporal, nonseparable, Matérn-like covariance function for $C_Y(\mathbf{h}; \tau)$, $\mathbf{h} \in \mathbb{R}^d$, $\tau = 0, \pm 1, \pm 2, \dots$, which they note is also valid when $-\infty < \tau < \infty$.

Perhaps surprisingly, it is possible to start with a valid spatio-temporal covariance function, $\{C_Y(\mathbf{h}; \tau) : \mathbf{h} \in \mathbb{R}^d, \tau \in \mathbb{R}\}$ and, under regularity conditions, obtain a spatio-temporal process Y that satisfies the stochastic IDE (6.63). Storvik, Frigessi, and Hirst (2002) show that (i) if $f_{Y,\tau}(\boldsymbol{\omega})$ exists for any $\tau \in \mathbb{R}$; (ii) if we can write

$$f_{Y,\tau}(\boldsymbol{\omega}) = f_{Y,0}(\boldsymbol{\omega})\{f(-\boldsymbol{\omega})\}^{|\tau|}, \quad \tau \in \mathbb{R}, \quad (6.65)$$

where the function $\{f(\omega): \omega \in \mathbb{R}^d\}$ has an inverse Fourier transform $m(\mathbf{h})$, $|f(\cdot)| < 1$, $f_{Y,0}(\cdot) > 0$, and $\int f_{Y,0}(\omega) d\omega < \infty$; and (iii) if we define

$$f_\eta(\omega) \equiv f_{Y,0}(\omega)\{(1 - f(\omega)f(-\omega))\} \quad (6.66)$$

and let $\eta_t(\cdot)$ be a temporally white and spatially colored stochastic process with spectral density $f_\eta(\cdot)$; then there exists a stochastic process $Y(\mathbf{s}; t)$ that satisfies (6.63). They then use this result to give a stochastic IDE representation for a process that has covariance function given by one of the nonseparable examples due to Cressie and Huang (1999).

In conclusion, it is possible to investigate which spatio-temporal covariance functions have a natural, physical interpretation as a stochastic IDE. Essentially, there has to be a “well behaved” function $f(\cdot)$ in (6.65), and the error process $\eta_t(\cdot)$ has to be chosen to satisfy (6.66); in this case, a stochastic IDE can be recovered.

6.3.2 The Diffusion-Injection Equation, Revisited

In this book, we encourage spatio-temporal statistical modelers to build their models from physical/chemical/biological/economic/etc. relationships and to use them to *motivate* various levels of the HM. Here, we choose a homogeneous reaction-diffusion PDE given by (6.2); we give further examples in Sections 6.3.1 and 7.2.4. In what is to follow, the PDE (6.2) is discretized, both temporally and spatially, resulting in the equation, $\mathbf{Y}_{t+\Delta_t} \simeq \mathbf{M}\mathbf{Y}_t$, where the propagator matrix $\mathbf{M}$ is structured (tridiagonal and very sparse). Then the resulting *stochastic* difference equation will be of the VAR(1) type given by (3.98); see also (7.46) and Wikle (2003b).

The PDE (6.2) has a first-order partial derivative in time and a second-order partial derivative in (one-dimensional) space. There is some choice involved in replacing the first-order derivative with a forward or a backward difference and replacing the second-order derivative with a forward, centered, or backward difference; which of these one chooses will make a (hopefully small) difference to the solution meant to approximate the PDE.

Replacing the first-order derivative with a forward difference and the second-order derivative with a centered difference, (6.2), is approximated as

$$\frac{Y(\mathbf{s}; t + \Delta_t) - Y(\mathbf{s}; t)}{\Delta_t} = \beta \left\{ \frac{Y(\mathbf{s} + \Delta_s; t) - 2Y(\mathbf{s}; t) + Y(\mathbf{s} - \Delta_s; t)}{\Delta_s^2} \right\} - \alpha Y(\mathbf{s}; t),$$

which reduces to

$$Y(\mathbf{s}; t + \Delta_t) = \theta_1 Y(\mathbf{s}; t) + \theta_2 Y(\mathbf{s} + \Delta_s; t) + \theta_2 Y(\mathbf{s} - \Delta_s; t), \quad (6.67)$$

where $\theta_1 \equiv (1 - \alpha\Delta_t - 2\beta\Delta_t/\Delta_s^2)$, $\theta_2 \equiv \beta\Delta_t/\Delta_s^2$, and recall that $\alpha > 0$ and $\beta > 0$. We shall see below that (6.67) approximates the PDE (6.2) well, provided that $\alpha\Delta_t < 1$ and $2\beta\Delta_t/\Delta_s^2 < 1$.

From (6.67), the next (future) value of Y at s is related to current values at s and neighboring spatial locations. Now let $D_s \equiv \{s_0, \dots, s_{n+1}\}$, where $s_j = s_0 + j \cdot \Delta_s$; $j = 0, 1, \dots, n+1$. The locations s_0 and s_{n+1} are spatial boundary locations. Then,

$$\begin{aligned} Y(s_1; t + \Delta_t) &= \theta_1 Y(s_1; t) + \theta_2 Y(s_2; t) + \theta_2 Y(s_0; t) \\ Y(s_2; t + \Delta_t) &= \theta_1 Y(s_2; t) + \theta_2 Y(s_3; t) + \theta_2 Y(s_1; t) \\ &\vdots \\ Y(s_n; t + \Delta_t) &= \theta_1 Y(s_n; t) + \theta_2 Y(s_{n+1}; t) + \theta_2 Y(s_{n-1}; t). \end{aligned}$$

Define $\mathbf{Y}_t \equiv (Y(s_1; t), \dots, Y(s_n; t))'$ and $\mathbf{Y}_t^{(b)} \equiv (Y(s_0; t), Y(s_{n+1}; t))'$, and so the equations given just above can be written as the (nonstochastic) vector difference equation,

$$\mathbf{Y}_{t+\Delta_t} = \mathbf{M}\mathbf{Y}_t + \mathbf{M}^{(b)}\mathbf{Y}_t^{(b)}, \quad (6.68)$$

where $\mathbf{M}$ is an $n \times n$ tridiagonal propagator matrix, and $\mathbf{M}^{(b)}$ is an $n \times 2$ boundary propagator matrix:

$$\mathbf{M} \equiv \begin{bmatrix} \theta_1 & \theta_2 & 0 & \cdots & \cdots & 0 \\ \theta_2 & \theta_1 & \theta_2 & \cdots & \cdots & 0 \\ 0 & \theta_2 & \theta_1 & \ddots & & \vdots \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \vdots & & \ddots & \theta_1 & \theta_2 \\ 0 & 0 & 0 & \cdots & \theta_2 & \theta_1 \end{bmatrix}, \quad \mathbf{M}^{(b)} \equiv \begin{bmatrix} \theta_2 & 0 \\ 0 & 0 \\ \vdots & \vdots \\ 0 & \theta_2 \end{bmatrix}. \quad (6.69)$$

Henceforth, let $D_t = \{0, 1, 2, \dots\}$, in units of Δ_t . Then, from a given initial condition $\mathbf{Y}_0$ and boundary conditions $\{\mathbf{Y}_t^{(b)}: t = 0, 1, \dots\}$, we can obtain a numerical solution to the difference equation (6.67), which is an approximation to the PDE (6.2). The sparseness of $\mathbf{M}$ helps with computation, but restrictions on Δ_t , Δ_s , α , and β are still needed to avoid unstable solutions of the difference equation. These restrictions ensure that the maximum absolute eigenvalue of $\mathbf{M}$ is less than 1, and they are

$$\alpha\Delta_t < 1 \quad \text{and} \quad 2\beta\Delta_t/\Delta_s^2 < 1.$$

For example, $\alpha = 1$, $\beta = 20$, $\Delta_t = .01$, and $\Delta_s = 1$ satisfy these inequalities (see Figure 6.1a).

Now add a mean-zero, random error component, δ_{t+1} , to the right-hand side of (6.68):

$$\mathbf{Y}_{t+1} = \mathbf{M}\mathbf{Y}_t + \mathbf{M}^{(b)}\mathbf{Y}_t^{(b)} + \delta_{t+1}, \quad t = 0, 1, \dots, \quad (6.70)$$

where $\{\delta_t\}$ are *iid*. This is a *stochastic* spatio-temporal difference equation [which, following Whittle (1986), might be called a *diffusion-injection difference equation*] corresponding to the stochastic PDE (6.1). Since the boundary term, $\mathbf{M}^{(b)}\mathbf{Y}_t^{(b)}$, is given, this stochastic difference equation can be recognized as a multivariate first-order autoregressive model, or VAR(1) model, already considered in Section 3.4.7. Assuming temporal stationarity of the covariance function, we see from a generalization of equation (3.101) that for any nonnegative integer k the $n \times n$ covariance matrix, $\Sigma_Y^{(k)} \equiv \text{cov}(\mathbf{Y}_{t+k}, \mathbf{Y}_t)$, satisfies

$$\Sigma_Y^{(k)} = \mathbf{M}^k \Sigma_Y, \quad k = 0, 1, 2, \dots, \quad (6.71)$$

where $\Sigma_Y \equiv \Sigma_Y^{(0)} = \text{var}(\mathbf{Y}_t)$. Consider the spatio-temporal stationary covariance function $C(h; \tau)$ in (6.3), obtained from the stochastic PDE (6.1), where $h > 0$, $\tau > 0$. For a given temporal lag τ , choose k_τ to be the integer part of τ/Δ_t . Furthermore, for a given spatial lag h , choose ℓ_h to be the integer part of h/Δ_s , and since D_s is fixed, n is $O(1/\Delta_s)$. Then the $(i, i + \ell_h)$ th element of $\Sigma_Y^{(k_\tau)}$ should approximate $C(h, \tau)$, as we now discuss.

Define $\Sigma_\delta \equiv \text{var}(\delta_t)$, which is the $n \times n$ covariance matrix of the random errors, and evaluate $\text{var}(\mathbf{Y}_{t+1})$ in (6.70):

$$\Sigma_Y = \mathbf{M}\Sigma_Y\mathbf{M}' + \Sigma_\delta. \quad (6.72)$$

For $\mathbf{M}$ and Σ_δ known, we are able to solve for Σ_Y using (6.72); hence, from (6.71), we obtain $\{\Sigma_Y^{(k)}: k = 0, 1, \dots\}$, which fully specifies the discrete-time, discrete-space spatio-temporal covariance function. The solution for Σ_Y is given by

$$\text{vec}(\Sigma_Y) = \{\mathbf{I} - \mathbf{M} \otimes \mathbf{M}\}^{-1} \text{vec}(\Sigma_\delta), \quad (6.73)$$

where recall that $\text{vec}(\cdot)$ is the operator on a matrix that strings its columns one underneath the other, and $\otimes$ denotes the Kronecker product such that $\mathbf{A} \otimes \mathbf{B}$ is a matrix whose (i, j) th block is $a_{ij}\mathbf{B}$. Consequently, $\text{vec}(\Sigma_Y)$ and $\text{vec}(\Sigma_\delta)$ are $n^2 \times 1$ vectors, and $\mathbf{I}$ and $\mathbf{M} \otimes \mathbf{M}$ above are $n^2 \times n^2$ matrices. Thus, the spatio-temporal covariance of the stochastic difference equation (that approximates the stochastic PDE) is completely determined by θ_1 , θ_2 , and Σ_δ (more precisely, by Δ_t , Δ_s , α , β , and $\text{cov}(\delta_t)$).

Recall that $D_s = \{s_0, s_1, \dots, s_n, s_{n+1}\}$, where $s_j = s_0 + j \cdot \Delta_s$; $j = 0, 1, \dots, n+1$. Let $\{Y_t(\mathbf{s}_j): j = 0, 1, \dots, n+1, t = 0, 1, \dots\}$ be the

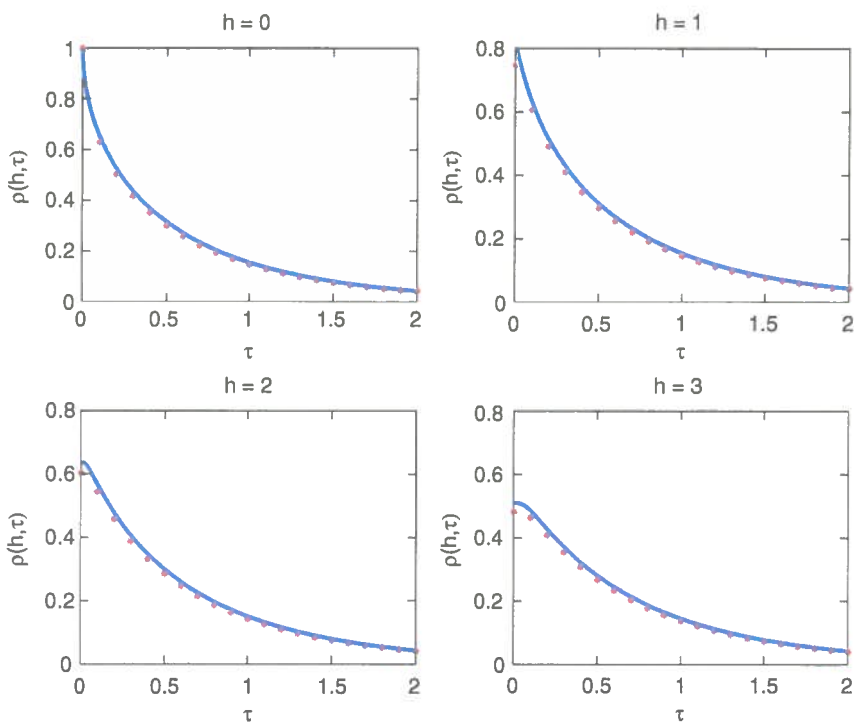


Figure 6.16 Comparison of the discretized correlation function given in (6.75) with the continuous correlation function given in (6.3), as a function of temporal lag τ for several choices of spatial lag h . Here $\alpha = 1$, $\beta = 20$, $\Delta_s = 1$, and $\Delta_t = 0.01$. The red dots display discretized correlation values obtained from (6.75) at time intervals of $10\Delta_t = 0.1$, and the blue line displays the continuous correlation function obtained from (6.3).

spatio-temporal process that solves the discrete-time, discrete-space stochastic difference equation (6.70). Then,

$$\text{cov}(Y_t(s_j), Y_{t+k}(s_{j+\ell})) = (\Sigma_Y^{(k)})_{j,j+\ell}, \quad (6.74)$$

where $\Sigma_Y^{(k)}$ is evaluated from (6.71) and (6.73). Now let $j = [n/2]$, and recall that $\ell_h = [h/\Delta_s]$ and $k_\tau = [\tau/\Delta_t]$ ($[x]$ denotes the integer part of x), where $\Delta_t = .01$ and $\Delta_s = 1$. Figure 6.16 shows plots of

$$\tilde{\rho}(h; \tau) \equiv (\Sigma_Y^{(k_\tau)})_{j,j+\ell_h} / \{(\Sigma_Y)_{jj} \cdot (\Sigma_Y)_{j+\ell_h,j+\ell_h}\}^{1/2}, \quad (6.75)$$

as functions of τ for several choices of h . On those plots is superimposed the continuous spatio-temporal correlation function (6.3) shown in Figure 6.3. The agreement between (6.74) and (6.3) is very good and will get better for smaller Δ_t and Δ_t/Δ_s^2 .

The approach given for approximating (6.1) with (6.70) is quite general; see Section 7.2.4 for further examples. Thus, stochastic PDEs with intractable solutions can be replaced with VAR processes that are amenable to statistical analysis. Furthermore, the VAR parameters (here θ_1, θ_2) are scientifically interpretable by relating them back to the approximated PDE.

6.4 TIME SERIES OF SPATIAL PROCESSES

In Section 6.3, we saw how a continuous-time model can be approximated upon introducing a small time-increment, Δ_t . There, $D_t = [0, \infty)$ is replaced with $D_t = \{0, 1, 2, \dots\}$, in units of Δ_t .

Henceforth, in this section, we choose $D_t = \{0, 1, 2, \dots\}$, namely a discrete-time model. Since the time index is discrete, the spatio-temporal process can be written as

$$\begin{aligned} Y_0(\cdot) &\equiv \{Y(\mathbf{s}; 0) : \mathbf{s} \in D_s\}, \\ Y_1(\cdot) &\equiv \{Y(\mathbf{s}; 1) : \mathbf{s} \in D_s\}, \\ &\dots \\ Y_t(\cdot) &\equiv \{Y(\mathbf{s}; t) : \mathbf{s} \in D_s\}, \end{aligned} \tag{6.76}$$

which could be generalized (but will not be in this section) by allowing the spatial index set D_s in (6.76) to depend on the time point t . That is, in (6.76), we have approximated the spatio-temporal process, $Y(\cdot; \cdot)$, as a *time series of spatial processes*:

$$\{Y_t(\cdot) : t = 0, 1, \dots\}. \tag{6.77}$$

We note here and below that one could take the conjugate point of view, namely that the spatio-temporal process could be thought of as a spatial process of time series. That is, think of the spatio-temporal process as a multivariate spatial process, where the multivariate dependencies are in fact inherited from temporal dependencies that may vary with spatial location (e.g., Bennett, 1979; Sansó, Schmidt, and Nobre, 2008).

In the subsections that follow, we consider both continuous-space and discrete-space models for (6.77).

6.4.1 Continuous Spatial Index (Geostatistical Processes)

Formally speaking, the continuous-spatial-index case is defined by $|D_s| > 0$, where recall that for $A \subset \mathbb{R}^d$, $|A| \equiv \int_A d\mathbf{s}$ is the volume (or Lebesgue measure) of A . In the purely spatial context, models for $\{Y(\mathbf{s}) : \mathbf{s} \in D_s\}$ have been called *geostatistical processes* (Cressie, 1993, Chapter 2). Thus, our interest here is in time series of geostatistical processes given by (6.77).

In Chapter 3, an example of a dynamical model for a time series of random variables $\{Y_t : t = 0, 1, \dots\}$ was given as

$$Y_t = \mathcal{M}_t(Y_{t-1}) + \delta_t, \quad t = 0, 1, \dots, \quad (6.78)$$

where $\mathcal{M}_t(\cdot)$ is a possibly nonlinear function that may depend on t , δ_t is an additive error term independent of Y_{t-1} , and $\{\delta_t : t = 0, 1, \dots\}$ are mutually independent. Using (6.78) as motivation, we could define a time series of geostatistical processes through

$$Y_t(\mathbf{s}) = \mathcal{M}_t(\mathbf{s}, Y_{t-1}(\cdot)) + \delta_t(\mathbf{s}), \quad \mathbf{s} \in D_s, \quad t = 0, 1, \dots, \quad (6.79)$$

where the spatial process $\delta_t(\cdot)$ is independent of the spatial process $Y_{t-1}(\cdot)$, and $\{\delta_t(\cdot) : t = 0, 1, \dots\}$ are mutually independent processes. For example, Wikle and Cressie (1999) considered models of the form of (6.79), where $\delta_t(\cdot)$ was a spatially colored error process with mean zero, and $\mathcal{M}_t$ did not depend on t . That is, they considered

$$Y_t(\mathbf{s}) = \mathcal{M}(\mathbf{s}, Y_{t-1}(\cdot)) + \delta_t(\mathbf{s}),$$

where

$$\mathcal{M}(\mathbf{s}, f(\cdot)) \equiv \int_{\mathbb{R}^d} m(\mathbf{s}, \mathbf{x}) f(\mathbf{x}) d\mathbf{x}, \quad \mathbf{s} \in D_s, \quad (6.80)$$

for $f(\cdot)$ any function in $\mathbb{R}^d$ such that the integral on the right-hand side of (6.80) exists. The weight function $m(\mathbf{s}, \mathbf{x})$ in (6.80) controls which parts of, and by how much, the process $Y_{t-1}(\cdot)$ influences the current value $Y_t(\mathbf{s})$. Notice that $\{\delta_t(\cdot) : t = 0, 1, \dots\}$ is a time series of error processes that are independent in time; however, each error process can exhibit spatial dependence. Figure 6.17 shows a schematic diagram of the dynamical structure found in (6.79) and (6.80). The diagram is meant to show intuitively how dependencies are transmitted through time and space; in fact, it could be formally interpreted as an acyclic-directed graph (Section 2.4). Further discussion of dynamical models like (6.80) can be found in Section 7.2.5.

There is a lot of flexibility in (6.79), to generate nonstationary spatio-temporal processes; that is, the action of the function $\mathcal{M}_t(\mathbf{s}, \cdot)$ may vary with $\mathbf{s}$ and t . In order to visualize some possibilities, we consider the same spatio-temporal domain as in Figure 6.8, and we divide $D_s = [0, 40]$ and $D_t = [0, 0.8]$ into 40 (i.e., $\Delta_s = 1$) and 80 (i.e., $\Delta_t = .01$) intervals, respectively. That is, we are considering a fine grid on (continuous one-dimensional space) $\times$ (time) that has $40 \times 80 = 3200$ pixels. Then the continuous function $\mathcal{M}_t(\mathbf{s}, \cdot)$; $\mathbf{s} \in D_s$, $t \in D_t$, can be discretized onto the fine grid by a sequence of 40×40 matrices, $\{\mathbf{M}_t : t = \Delta_t, 2\Delta_t, \dots, 80\Delta_t\}$.

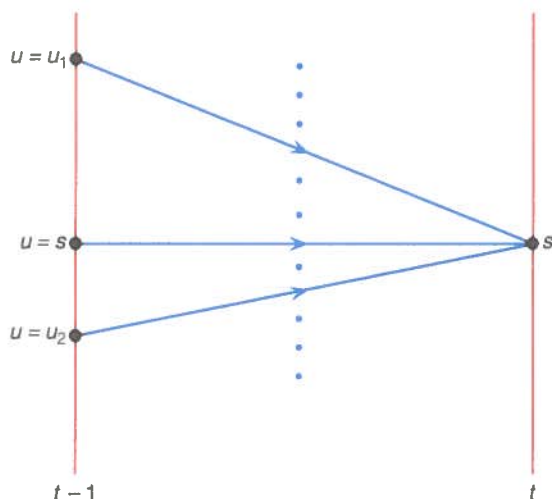


Figure 6.17 Schematic diagram (directed graph) of the dynamic structure found in (6.79) and (6.80); $D_s \subset \mathbb{R}^1$.

For example, consider,

$$\mathbf{M}_t \equiv h_t \cdot \mathbf{M} \equiv h_t \cdot \begin{bmatrix} \mathbf{M}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_2 \end{bmatrix}, \quad (6.81)$$

where the 20×20 matrices $\mathbf{M}_1$ and $\mathbf{M}_2$ are given by

$$\mathbf{M}_1 \equiv \begin{bmatrix} .90 & .01 & 0 & \cdots & 0 \\ .01 & .90 & 0 & \cdots & 0 \\ 0 & 0 & \ddots & & \vdots \\ \vdots & \vdots & & .90 & .01 \\ 0 & 0 & \cdots & .01 & .90 \end{bmatrix}, \quad \mathbf{M}_2 \equiv \begin{bmatrix} .20 & .01 & 0 & \cdots & 0 \\ .01 & .20 & 0 & \cdots & 0 \\ 0 & 0 & \ddots & & \vdots \\ \vdots & \vdots & & .02 & .01 \\ 0 & 0 & \cdots & .01 & .20 \end{bmatrix}$$

and

$$h_t \equiv \begin{cases} 1, & \text{if } 1 \leq t \leq 30\Delta_t, \\ -1, & \text{if } 31\Delta_t \leq t \leq 60\Delta_t, \\ 1, & \text{if } 61\Delta_t \leq t \leq 80\Delta_t. \end{cases}$$

Clearly, in $D_s \times D_t$, there are six distinct regions, corresponding to weak/strong spatial dependence $\times$ positive/negative/positive temporal dependence. In spite of these heterogeneities, it is possible to define a $Y_t(\cdot)$ so that it has the same spatial covariance structure for each t , as we shall now demonstrate.

In terms of the grid of 3200 pixels, the geostatistical process $Y_t(\cdot)$ at any given t becomes $\mathbf{Y}_t \equiv (Y_t(1), \dots, Y_t(40))'$, and the error process $\delta_t(\cdot)$ becomes $\boldsymbol{\delta}_t \equiv (\delta_t(1), \dots, \delta_t(40))'$; $t = .01, .02, \dots, .8$. Then, in terms of vectors and matrices, the time series of geostatistical processes can be written in its discretized form as

$$\mathbf{Y}_t = \mathbf{M}_t \mathbf{Y}_{t-1} + \boldsymbol{\delta}_t, \quad (6.82)$$

where we define

$$\text{var}(\mathbf{Y}_t) \equiv \boldsymbol{\Sigma}_{Y,t} \quad \text{and} \quad \text{var}(\boldsymbol{\delta}_t) \equiv \boldsymbol{\Sigma}_{\delta,t}.$$

Clearly, (6.82) could also be used to define a time series of lattice processes, but in this example we have in mind a geostatistical process for which Δ_t and Δ_t/Δ_s^2 are small, such as for the diffusion-injection model considered in Section 6.3.2.

From (6.82), we have a generalization of (6.71) and (6.72):

$$\boldsymbol{\Sigma}_{Y,t} = \mathbf{M}_t \boldsymbol{\Sigma}_{Y,t-1} \mathbf{M}_t' + \boldsymbol{\Sigma}_{\delta,t},$$

and

$$\text{cov}(\mathbf{Y}_{t+\tau}, \mathbf{Y}_t) = \mathbf{M}_{t+\tau} \mathbf{M}_{t+\tau-1} \cdots \mathbf{M}_{t+1} \boldsymbol{\Sigma}_{Y,t}.$$

In the example, we choose

$$\boldsymbol{\Sigma}_{Y,t} \equiv \boldsymbol{\Sigma} \equiv (\exp(-|i-j|/10)) + 4 \cdot \mathbf{I}_{40}, \quad (6.83)$$

where the expression in parentheses on the right-hand side of (6.83) gives the (i, j) th entry of the 40×40 matrix. Notice that (6.83) does not depend on t . Consequently, if the error process is generated according to $\boldsymbol{\delta}_t \sim \text{Gau}(\mathbf{0}, \boldsymbol{\Sigma}_{\delta,t})$, where

$$\boldsymbol{\Sigma}_{\delta,t} \equiv \boldsymbol{\Sigma} - \mathbf{M}_t \boldsymbol{\Sigma} \mathbf{M}_t', \quad t = \Delta_t, 2\Delta_t, \dots,$$

and the initial value in the time series of the geostatistical processes is generated according to $\mathbf{Y}_0 \sim \text{Gau}(\mathbf{0}, \boldsymbol{\Sigma})$, then the recursion given by (6.82) results in a spatio-temporal process, $\mathbf{Y}_{\Delta_t}, \mathbf{Y}_{2\Delta_t}, \dots$, with homogeneous $\text{var}(\mathbf{Y}_{\Delta_t}) = \text{var}(\mathbf{Y}_{2\Delta_t}) = \cdots = \boldsymbol{\Sigma}$. This gives a valid spatio-temporal process because $\boldsymbol{\Sigma} - \mathbf{M}_t \boldsymbol{\Sigma} \mathbf{M}_t'$ is positive-definite, and this occurs if the eigenvalues of $\mathbf{M}_t \boldsymbol{\Sigma} \mathbf{M}_t'$ are all strictly less than 1. The model (6.81) satisfies this condition. Figure 6.18 shows three realizations of the spatio-temporal process, where the six regions we referred to earlier are clearly visible.

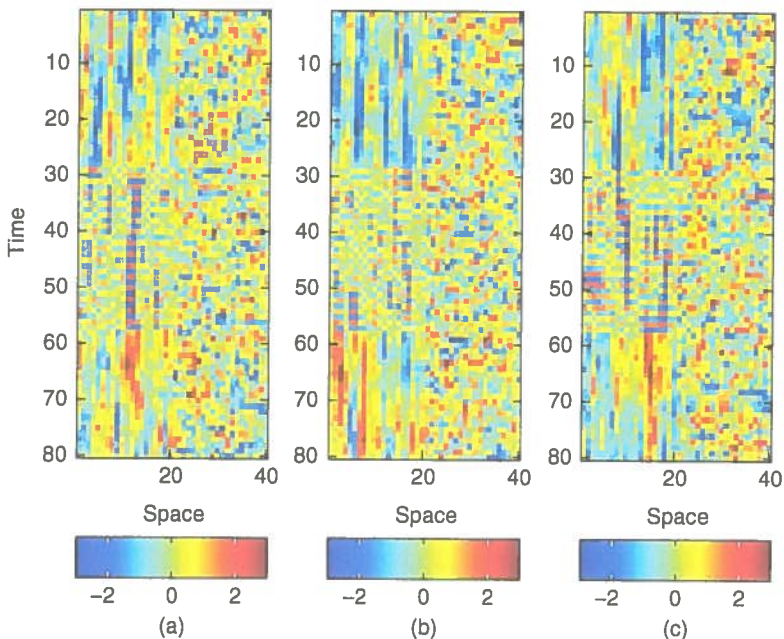


Figure 6.18 Three independent realizations of a non-stationary spatio-temporal process, constructed from (6.81) and (6.82). In the figure, there are six spatio-temporal domains that exhibit different behavior: $\{1, \dots, 20\} \times \{1, \dots, 30\}$; $\{1, \dots, 20\} \times \{31, \dots, 60\}$; $\{1, \dots, 20\} \times \{61, \dots, 80\}$; $\{21, \dots, 40\} \times \{1, \dots, 30\}$; $\{21, \dots, 40\} \times \{31, \dots, 60\}$; $\{21, \dots, 40\} \times \{61, \dots, 80\}$.

The cross-covariances are generally nonstationary. In Figure 6.19, we show various $\text{corr}(Y(s; t), Y(x; r))$, obtained from its covariance-matrix form:

$$\text{cov}(\mathbf{Y}_{t+\tau}, \mathbf{Y}_t) = \mathbf{M}_{t+\tau} \mathbf{M}_{t+\tau-1} \cdots \mathbf{M}_{t+1} \boldsymbol{\Sigma}, \quad (6.84)$$

where $\mathbf{M}_t$ is given by (6.81) and $\boldsymbol{\Sigma}$ is given by (6.83).

From an alternative perspective, a spatial process of time series might be modeled through a temporally stationary covariance function $C_s(\tau)$, $\tau = 0, \pm 1, \pm 2, \dots$, where here the subscript indexes the *location* of the time series. Then the theorem of Cressie and Huang (1999) that was stated in Section 6.1.6 could be used to construct a coherent spatio-temporal covariance for $s \in D_s$ and $t = 0, 1, 2, \dots$.

Our focus in this subsection has been on dynamical models of geostatistical processes, even if practical considerations force a discretization of D_s into a collection of small finite elements. Philosophically, the spatial index is continuous, reflected by the possibility that the finite elements might be made larger or smaller, *and* that the geostatistical process $Y_t(\cdot)$ will not be observed at every spatial location $s \in D_s$. One of the opening examples of this chapter was a spatio-temporal process of air pressure anywhere in the earth's atmosphere and at any time. Then, for this example, the spatial index set D_s is clearly

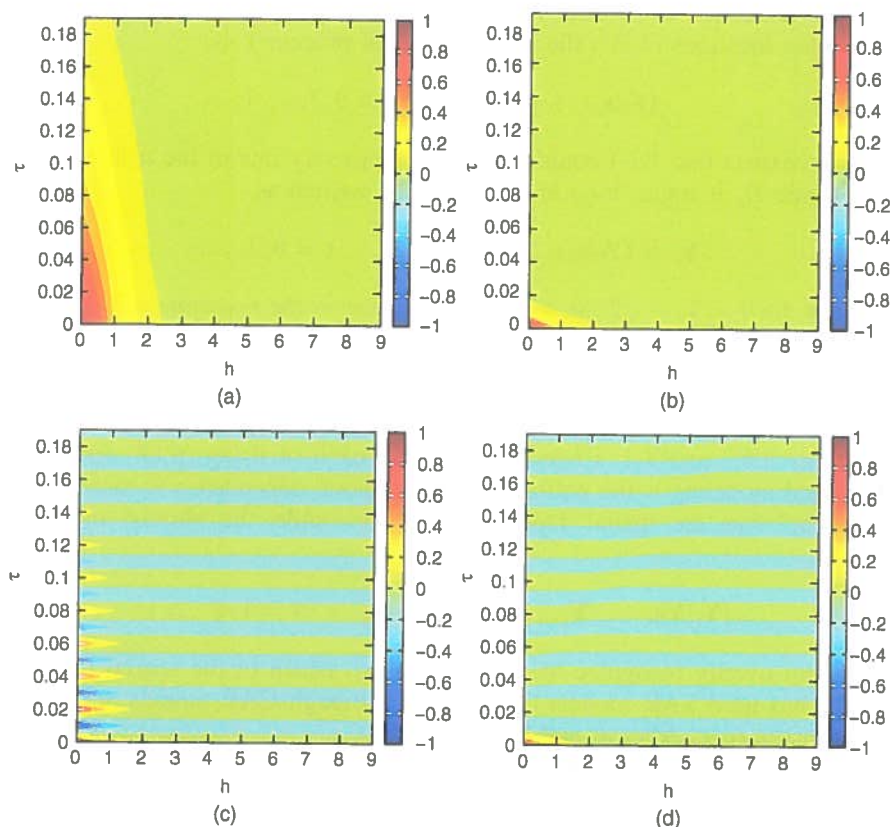


Figure 6.19 (a) Contour plot of $\text{corr}(Y(1; 1), Y(1 + h; 1 + \tau))$, exhibiting strong positive correlation. (b) Contour plot of $\text{corr}(Y(21; 1), Y(21 + h; 1 + \tau))$, exhibiting weak positive correlation. (c) Contour plot of $\text{corr}(Y(1; 31), Y(1 + h; 31 + \tau))$, exhibiting strong oscillatory correlation. (d) Contour plot of $\text{corr}(Y(21; 31), Y(21 + h; 31 + \tau))$, exhibiting weak oscillatory correlation.

continuous. Another opening example was based on yearly breast-cancer incidence in the 88 counties of Ohio, for which D_s is clearly discrete and is an instance of a time series of *lattice* processes.

6.4.2 Discrete Spatial Index (Lattice Processes)

When D_s is finite or countable, its Lebesgue measure is zero; that is, $|D_s| = 0$. An example of a countable $D_s \subset \mathbb{R}^d$ is the d -dimensional grid of integers, $\{0, 1, \dots\}^d$. In practice, such grids are usually truncated and made finite. For the rest of this subsection, we assume

$$D_s = \{s_1, s_2, \dots, s_n\}, \quad (6.85)$$

a *finite* collection of locations in space. Notice that we are not interested in any other locations in $\mathbb{R}^d$; the spatio-temporal process Y is

$$\{Y_t(\mathbf{s}_i): i = 1, \dots, n; t = 0, 1, \dots\}, \quad (6.86)$$

and we expect that $Y_t(\cdot)$ could be observed at every one of the n locations in D_s . Since D_s is finite, $Y_t(\cdot)$ in (6.77) can be written as

$$\mathbf{Y}_t \equiv (Y_t(\mathbf{s}_1), \dots, Y_t(\mathbf{s}_n))', \quad t = 0, 1, \dots \quad (6.87)$$

That is, for $t = 1, \dots, T$, (6.87) can be written as the n -variate time series,

$$\mathbf{Y}_0, \mathbf{Y}_1, \dots, \mathbf{Y}_T. \quad (6.88)$$

Additional discussion of multivariate time series models can be found in Sections 3.4.7 and 7.4. Underlying the approach of using (6.88) for spatio-temporal modeling is the notion that the temporal dependence is more natural to model than the spatial dependence. For example, the Markov model for (6.88),

$$[\mathbf{Y}_t | \mathbf{Y}_0, \dots, \mathbf{Y}_{t-1}] = [\mathbf{Y}_t | \mathbf{Y}_{t-1}], \quad t = 1, 2, \dots, \quad (6.89)$$

does not overtly recognize the spatial context. Much of the spatial modeling of $\mathbf{Y}_t$ has used SAR models (see below), although CAR models are equally appropriate (e.g., Allcroft and Glasbey, 2003).

Modeling through (6.88) is in contrast to modeling $\{Y_t(\mathbf{s}_i)\}$ as a spatial process of time series, namely $\{\mathbf{Y}(\mathbf{s}_i): i = 1, \dots, n\}$, where $\mathbf{Y}(\mathbf{s}_i) \equiv (Y_0(\mathbf{s}_i), \dots, Y_T(\mathbf{s}_i))'$; see Sansó, Schmidt, and Nobre (2008). One could also think of the spatio-temporal dependence in terms of a $(T + 1)$ -variate spatial lattice process [using, for example, models such as those presented by Sain and Cressie (2007)].

Vector Autoregressive Models

Without simplifying assumptions, the number of parameters specifying the distribution of (6.88) can be enormous. The model (6.89), say, uses the temporal dimension to achieve a more parsimonious model. By recognizing spatial “neighborhood” relationships, we can use the spatial dimension to achieve further parsimony. These ideas are all discussed in Chapter 7, and we give a simple example in the next paragraph.

A special case of (6.89) is the VAR(1) model of dimension n (Section 3.4.7):

$$\mathbf{Y}_t = \mathbf{M}\mathbf{Y}_{t-1} + \boldsymbol{\eta}_t, \quad (6.90)$$

where, in its full generality, $\mathbf{M}$ has n^2 parameters and $\boldsymbol{\Sigma}_\eta = \text{var}(\boldsymbol{\eta}_t)$ has $O(n^2)$ parameters. The spatial context can be used to reduce the number of parameters drastically. For example, suppose that we assume the (i, j) th entry of $\mathbf{M}$, $(\mathbf{M})_{ij} = 0$, unless $\|\mathbf{s}_i - \mathbf{s}_j\| \leq h$. Then the current value at $\mathbf{s}_i$ is related

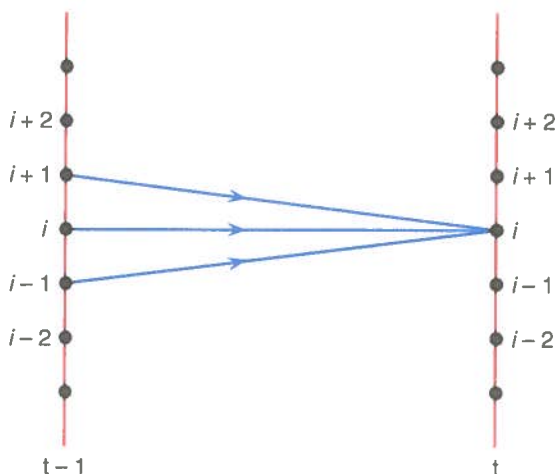


Figure 6.20 Schematic diagram (directed graph) of the dynamic structure found in (6.90) when $\mathbf{M}$ is defined spatially.

to those immediate-past values at $\mathbf{s}_i$ and nearby $\mathbf{s}_j$ (within a radius of h). Figure 6.20 shows a schematic diagram (i.e., a directed graph) of the dynamic structure in (6.90) when $\mathbf{M}$ is defined spatially, here in $\mathbb{R}^1$ with D_s a subset of the integers and $h = 1$. Thus, rather than $\mathbf{M}$ being made up of n^2 parameters, the parameter space has been reduced by making $\mathbf{M}$ sparse through spatial proximities of locations in D_s ; in Figure 6.20, $(\mathbf{M})_{ij} = 0$, unless $|i - j| \leq 1$. A similar modeling strategy that allows further reduction in the size of the parameter space would choose Σ_η to be sparse (a geostatistical-type model; see Section 4.1) or Σ_η^{-1} to be sparse (a lattice-type model; see Section 4.2). A complete discussion of parsimonious parameterization of models like (6.90) is given in Sections 7.2 and 7.3.

We remark that the model (6.90) refers to the entirety of locations; however, the model does not allow inference on an $(n+1)$ th location $\mathbf{s}$. Consequently, a spatio-temporal statistical model that is defined by an n -dimensional multivariate time series, such as (6.90), cannot handle the problem of off-site predictions.

The VAR(1) model given by (6.90) is a special case of the spatio-temporal autoregressive moving-average (STARMA) model. It is generally true that for these and other multivariate time series, the number of parameters can be potentially enormous, and an important skill of the modeler is to reduce drastically the size of the parameter space. We believe that this is best achieved through recognizing and preserving any known spatio-temporal interactions (e.g., from a PDE) for the underlying process $Y(\cdot; \cdot)$; see Sections 6.3 and 7.2.4. In the absence of such knowledge, other approaches, such as those described throughout Chapter 7, can be taken.

Space–Time Autoregressive Moving-Average (STARMA) Models

In Section 6.3, we have seen how a spatio-temporal stochastic PDE can be discretized and written as a multivariate time series, $\{\mathbf{Y}_t\}$, that can be modeled dynamically as a VAR(1) process. One could look for even more generality in the temporal domain, by assuming higher orders of autoregression as well as a moving-average type of dependence. This would correspond to higher-order temporal partial derivatives and nonspecific temporal dependence in the noise term $\delta(\cdot; \cdot)$ of the stochastic PDE.

Define the *spatio-temporal autoregressive moving-average (STARMA)* model (Ali, 1979; Pfeifer and Deutsch, 1980; and Cressie, 1993, p. 450) as

$$\mathbf{Y}_t = \sum_{k=0}^p \left(\sum_{j=1}^{\lambda_k} f_{kj} \mathbf{U}_{kj} \right) \mathbf{Y}_{t-k} + \sum_{\ell=0}^q \left(\sum_{j=1}^{\mu_\ell} g_{\ell j} \mathbf{V}_{\ell j} \right) \mathbf{W}_{t-\ell}, \quad t = 0, 1, \dots, \quad (6.91)$$

where $\{\mathbf{U}_{kj}\}$ and $\{\mathbf{V}_{\ell j}\}$ are known weight matrices; p and q are the orders of the autoregressive part and moving-average part, respectively; $\{f_{kj}\}$ and $\{g_{\ell j}\}$ are parameters of the model; and $\{\mathbf{W}_t\}$ are *iid* random vectors with mean $\mathbf{0}$ and covariance matrix Σ_w . In (6.91), the index j is used to denote substructures, although including them makes the number of parameters potentially enormous. Upon reparameterizing, we obtain

$$\mathbf{Y}_t = \sum_{k=0}^p \mathbf{B}_k \mathbf{Y}_{t-k} + \sum_{\ell=0}^q \mathbf{E}_\ell \mathbf{W}_{t-\ell}, \quad (6.92)$$

where, without loss of generality, $\Sigma_w = \sigma_w^2 \mathbf{I}$, and, for identifiability reasons, $\mathbf{B}_0$ has zero entries down the diagonal. We assume henceforth that $(\mathbf{I} - \mathbf{B}_0)$ is invertible.

The number of parameters in (6.92) is still very large. Consider several simple cases. First, $p = 0$ and $q = 0$ reduces (6.92) to a sequence of purely spatial processes without any temporal dependence linking them:

$$\mathbf{Y}_t = \mathbf{B}_0 \mathbf{Y}_t + \mathbf{E}_0 \mathbf{W}_t, \quad t = 0, 1, \dots$$

Upon rewriting this expression, we obtain,

$$\mathbf{Y}_t = (\mathbf{I} - \mathbf{B}_0)^{-1} \mathbf{E}_0 \mathbf{W}_t, \quad t = 0, 1, 2, \dots, \quad (6.93)$$

and since $\{\mathbf{W}_0, \mathbf{W}_1, \dots\}$ are mutually independent, we see that the time series $\{\mathbf{Y}_t\}$ given by (6.93) has no temporal dependence. When $\mathbf{E}_0 = \mathbf{I}$, these random vectors are *iid* mean-zero simultaneous autoregressions (SARs); see Section 4.2.5.

The second case is $p = 1$ and $q = 0$, and recall that $\mathbf{B}_0$ has all-zero diagonal entries. Then,

$$\mathbf{Y}_t = \mathbf{B}_0 \mathbf{Y}_t + \mathbf{B}_1 \mathbf{Y}_{t-1} + \mathbf{E}_0 \mathbf{W}_t, \quad t = 0, 1, \dots$$

Given $\mathbf{Y}_{t-1}$, $\mathbf{Y}_t$ has spatial statistical dependence that is of the SAR form. From Cressie (1993, p. 409), a SAR can be written as a CAR, which is a Markov random field with simple *conditional* probability dependencies (Section 4.2.5). These dependencies are captured by an undirected graph (see Section 4.2.2). The equation just above can be written equivalently as,

$$\begin{aligned} \mathbf{Y}_t &= (\mathbf{I} - \mathbf{B}_0)^{-1} \mathbf{B}_1 \mathbf{Y}_{t-1} + (\mathbf{I} - \mathbf{B}_0)^{-1} \mathbf{E}_0 \mathbf{W}_t \\ &\equiv \mathbf{M} \mathbf{Y}_{t-1} + \boldsymbol{\eta}_t, \end{aligned} \quad (6.94)$$

where $\mathbf{M} \equiv (\mathbf{I} - \mathbf{B}_0)^{-1} \mathbf{B}_1$ and $\{\boldsymbol{\eta}_t\}$ are *iid* with mean zero and $\text{var}(\boldsymbol{\eta}_t) = \sigma_w^2 \boldsymbol{\Sigma}_\eta (\mathbf{I} - \mathbf{B}_0)^{-1} \mathbf{E}_0 \mathbf{E}_0' (\mathbf{I} - \mathbf{B}_0')^{-1}$. The matrix $\mathbf{B}_0$ represents “instantaneous” spatial dependence (see below).

Another way to achieve the model in the same form, is the case $p = 1$, $q = 0$, and $\mathbf{B}_0 \equiv \mathbf{0}$. Then,

$$\mathbf{Y}_t = \mathbf{B}_1 \mathbf{Y}_{t-1} + \mathbf{E}_0 \mathbf{W}_t, \quad t = 0, 1, \dots,$$

which is equivalent to

$$\mathbf{Y}_t = \mathbf{M} \mathbf{Y}_{t-1} + \boldsymbol{\eta}_t, \quad (6.95)$$

where now $\mathbf{M} \equiv \mathbf{B}_1$, and $\{\boldsymbol{\eta}_t\}$ are *iid* with mean zero and $\text{var}(\boldsymbol{\eta}_t) = \sigma_w^2 \mathbf{E}_0 \mathbf{E}_0'$. This is also a VAR(1) model.

One might then ask, “What is the purpose of $\mathbf{B}_0$?” Figure 6.21 gives a schematic diagram of the type of spatio-temporal dependence in (6.94), for which D_s is a subset of the integers in $\mathbb{R}^1$ and $\mathbf{B}_0$ is tridiagonal. In Figure 6.21a, notice the three directed edges from sites $i + 1, i, i - 1$ at time $t - 1$ to site i at time t . Notice also the four undirected edges at time t between site i and its neighbors at sites $i + 2, i + 1, i - 1$, and $i - 2$ (corresponding to instantaneous spatial dependence); this could be considered to define a chain graph, with a combination of directed and undirected edges (e.g., Lauritzen, 1996). Compare this to Figure 6.20, which in fact gives a schematic diagram of the spatio-temporal dependence in (6.95), for which $\mathbf{B}_0 = \mathbf{0}$ (no instantaneous spatial dependence). If we multiply out $(\mathbf{I} - \mathbf{B}_0)^{-1} \mathbf{B}_1$, where recall that $\mathbf{B}_0$ is tridiagonal, we obtain $\mathbf{M}$ in (6.94) that is generally *not* sparse. The graphical structure of (6.94) where $\mathbf{M}$ is not sparse, is shown in Figure 6.21b; notice that there are directed edges from *all* sites at time $t - 1$ to site i at time t . Even though Figures 6.21a and 6.21b represent the same type of model, the parameterization required for Figure 6.21a is much simpler.

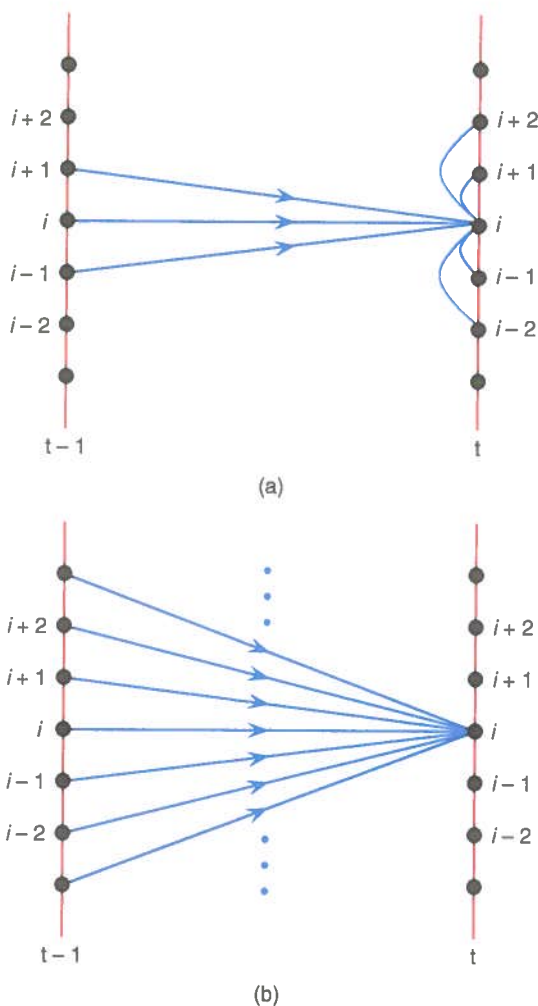


Figure 6.21 (a) Schematic diagram (chain graph) of the spatio-temporal dependence in (6.94), which includes instantaneous spatial dependence. (b) Schematic diagram (directed graph) of the spatio-temporal dependence structure with edges from all sites, which is equivalent to the spatio-temporal dependence exhibited in (a).

There are clearly a number of different ways to arrive at the same type of model. The difference between them lies in their parameterizations. When possible, we should look for a sparse but appropriate parameterization by considering the role each parameter plays in the physical model. One way to think of $\mathbf{B}_0$ is that it captures the variability at time steps much smaller than the unit of time specified for the autoregression. Small-temporal-scale dynamics, which may be important and unwise to ignore, are collected together into the matrix $\mathbf{B}_0$ that models *instantaneous spatial dependence* (Cressie, 1993,

p. 450). Thus, it is an approximation of dynamical structure running at time scales much smaller than the unit of time in the autoregression. Other interpretations are possible. Therefore, although the case of nonzero $\mathbf{B}_0$ in (6.94) and the case of $\mathbf{B}_0 = \mathbf{0}$ in (6.95) both lead to VAR(1) models, their interpretation is very different.

Now, consider the case $p = 1$, $q = 0$, $\mathbf{B}_0 = \mathbf{0}$, $\mathbf{E}_0 = \mathbf{I}$, and $\mathbf{B}_1 = \text{diag}\{b_1, \dots, b_n\}$. Then, the STARMA model reduces to

$$Y_t(\mathbf{s}_i) = b_i Y_{t-1}(\mathbf{s}_i) + W_t(\mathbf{s}_i), \quad i = 1, \dots, n, \quad t = 0, 1, \dots, \quad (6.96)$$

which is a collection of uncorrelated (in space) first-order autoregressive time series at each location $\{\mathbf{s}_i: i = 1, \dots, n\}$. In this case, prediction of values of the process at pixels where there are no data cannot “borrow strength” from observations at (nearby) pixels. Fortunately, the spatial context usually means that $\mathbf{B}_1$ is *not* a diagonal matrix, which makes this last case an unlikely one. Notice that if the “forcing matrix,” $\mathbf{E}_0$, is *not* the identity in the case discussed just above, $\mathbf{Y}_t$ is spatially dependent.

In the simple model (6.96), there is an opportunity to show the flexibility of conditional-probability modeling. Think of $\mathbf{b} \equiv (b_1, \dots, b_n)'$ in (6.96) as given; that is, the model (6.96) is

$$[\mathbf{Y}_t | \mathbf{Y}_{t-1}, \mathbf{b}] = \prod_{i=1}^n [Y_t(\mathbf{s}_i) | Y_{t-1}(\mathbf{s}_i), b_i], \quad t = 0, 1, \dots, \quad (6.97)$$

where the right-hand side specifies *spatial* (conditional) independence. Now think of the autoregressive coefficients $\mathbf{b}$ as a spatial process with distribution $[\mathbf{b}]$. Then, marginally,

$$[\mathbf{Y}_t | \mathbf{Y}_{t-1}] = \int \left\{ \prod_{i=1}^n [Y_t(\mathbf{s}_i) | Y_{t-1}(\mathbf{s}_i), b_i] \right\} [\mathbf{b}] d\mathbf{b}. \quad (6.98)$$

From the conditional model (6.98), $Y_t(\mathbf{s}_i)$ generally depends on all entries of $\mathbf{Y}_{t-1}$, which was not the case for the conditional model (6.96). Thus, through simple (conditional) models at successive levels of a hierarchy, complex (unconditional) structure can be captured. Wikle, Berliner, and Cressie (1998) used four levels of a hierarchical statistical model to model complex spatio-temporal dependence in meteorological data (Section 6.6). This hierarchical statistical approach to modeling spatio-temporal processes can be found in Chapter 7 and is applied to substantive problems in Chapter 9.

6.5 SPATIO-TEMPORAL POINT PROCESSES

Recall from Section 4.3 the definition of a spatial point process in a bounded subset D_s of $\mathbb{R}^d$ as a stochastic counting process defined on all Lebesgue

measurable subsets of D_s . It is a simple matter to extend the definition to a bounded subset, $D_{s,t}$, of $\mathbb{R}^d \times \mathbb{R}$. Let T denote the largest time in that subset; then there exists a bounded subset D_s of $\mathbb{R}^d$ such that $D_{s,t} \subset D_s \times [0, T]$. Hereafter, we write $D_{s,t} = D_s \times [0, T]$, and hence we can think about a spatio-temporal point process either as a point process in $\mathbb{R}^{d+1}$ (descriptive) or as a temporal process of spatial point processes (dynamical). Note that in some circumstances, a spatial point process of temporal processes might be a sensible model, but it is less likely and we do not pursue it here.

In this section, we consider only simple point processes (i.e., almost surely, either no event or a single event occurs at any point in space $\times$ time). Let $Z(A)$ denote the number of events in $A \subset D_s \times [0, T]$; if $Z(A)$ is (almost surely) finite for all such A , then $\{Z(A) : A \subset D_s \times [0, T]\}$ characterizes the point process. To emphasize the presence of spatial and temporal components, we sometimes refer to the point process as $Z(\cdot; \cdot)$. Indeed, from a dynamical perspective, $\{Z(\cdot; t) : t \in [0, T]\}$ and $\{Z_t(\cdot) : t = 0, 1, \dots\}$ represent, respectively, a continuously evolving and a discretely evolving class of spatial point processes.

Definitions for the (first-order) intensity, $\{\lambda(\mathbf{s}; t) : \mathbf{s} \in D_s, t \in [0, T]\}$, and higher-order moment measures follow in the same manner as in Section 4.3. In this brief presentation of spatio-temporal point processes, we go from the specific to the general. First, the *spatio-temporal (homogeneous) Poisson point process* Z has constant intensity λ^o , $Z(A) \sim \text{Poi}(\lambda^o|A|)$, where $|A|$ is the $(d+1)$ -dimensional volume of A , and for $A_1 \cap A_2 = \emptyset$, $Z(A_1)$ and $Z(A_2)$ are independent. The *inhomogeneous Poisson point process* is obtained by replacing λ^o with $\lambda(\cdot; \cdot)$, so that $Z(A) \sim \text{Poi}(\int_A \lambda(\mathbf{x}; r) d\mathbf{x}dr)$. For example, suppose

$$\lambda(\mathbf{s}; t) = \lambda^o(t), \quad \mathbf{s} \in D_s, t \in [0, T].$$

Then for fixed $t \in [0, T]$, $Z(\cdot; t)$ is a homogeneous Poisson process with constant (over space) intensity $\lambda^o(t)$.

It is straightforward to extend the definitions in Section 4.3 to define spatio-temporal versions of the *pair-correlation function*, the *K function*, and the *L function* (e.g., Gabriel and Diggle, 2009). It is also possible to extend the models in Section 4.3 to the spatio-temporal equivalent of *dependent (spatial Markov) point processes* and *marked point processes*. However, it is important to remember here that the time dimension is ordered and, where possible, point processes should respect the evolutionary nature of the phenomena they are modeling.

Spatial point processes generally do not have directionality associated with them (for an exception to this, see Section 4.3.5 where Directed Markov Point Processes are discussed). Hence, in the spatial setting, the notion of *conditional intensity*, conditional on “the past,” is not natural. Conditional intensities are attractive, because, as will be seen below, the point process’ likelihood function is straightforward to write down.

A simple point process can be characterized uniquely by its conditional intensity (Fishman and Snyder, 1976). The formal definition requires the

notions of σ -algebras, a filtration, and a martingale (e.g., Daley and Vere-Jones, 2003); in what follows, we give an intuitive presentation of the conditional intensity function in the spatio-temporal setting. Think about the entire “history,” $\mathcal{H}_t$, of a spatio-temporal point process up to time t . Then the conditional intensity, $\psi(\mathbf{s}; t)$, is the frequency with which events are expected to occur at $(\mathbf{s}; t)$, given $\mathcal{H}_t$. Formally,

$$\psi(\mathbf{s}; t) \equiv \lim_{\substack{|ds| \rightarrow 0 \\ dt \rightarrow 0}} \frac{E(Z(ds; dt) | \mathcal{H}_t)}{|ds| |dt|}, \quad (6.99)$$

where recall that ds is a small region located at $\mathbf{s}$, and likewise dt is a small time interval at t . We write the parameterized conditional intensity as $\psi(\mathbf{s}; t; \theta)$.

Let the events of Z be written as $\{(\mathbf{s}_i; t_i): i = 1, \dots, N\}$, where $N = Z(D_s \times [0, T])$. Then the loglikelihood function (up to an additive constant) is (e.g., Daley and Vere-Jones, 2003),

$$\sum_{i=1}^n \log \psi(\mathbf{s}_i; t_i; \theta) - \int_0^T \int_{D_s} \psi(\mathbf{x}; r; \theta) d\mathbf{x} dr. \quad (6.100)$$

This attractive expression belies the difficulty sometimes encountered in evaluating the integral in (6.100), as well as in interpreting the conditional intensity back in terms of the physical/chemical/biological/economic/etc. phenomenon originally being modeled. In its full generality, a spatio-temporal point process has a likelihood analogous to the rather complicated expression (4.186) in the spatial case; see the discussion surrounding this expression in Section 4.3.3. Yet, (6.100) is equally the (log)likelihood and appears to be in a much simpler form. Thus, the evolutionary aspect of the temporal component allows conceptual simplification. Of course, once a conditional intensity is fitted to the available data, it then needs to be interpreted properly, as a quantification of the *conditional* properties of the process.

A simple example of a joint specification of a spatio-temporal point process can be given through the inhomogeneous Poisson point process, which was defined at the beginning of this section. This process is characterized by the first-order intensity function, $\lambda(\mathbf{s}; t)$, but recall that it has a rather strong independence property for counts in nonintersecting sets A_1 and A_2 . However, some flexibility is achieved by making the first-order intensity random; that is, $\{\lambda(\mathbf{s}; t): \mathbf{s} \in D_s, t \in [0, T]\}$ is assumed to be a spatio-temporal *stochastic* process. This results in a spatio-temporal *Cox process* (Section 4.3.1) an example of which is the *log-Gaussian Cox process* (Møller, Syversveen and Waagepetersen, 1998). Here, $\lambda(\mathbf{s}; t) = \exp\{Y(\mathbf{s}; t)\}$, where $\{Y(\mathbf{s}; t): \mathbf{s} \in \mathbb{R}^d, t \in [0, T]\}$ is a Gaussian process whose behavior could be captured by any of the models discussed in Sections 6.1, 6.3, and 6.4.1 and in Chapter 7.

As an HM, the log Gaussian Cox process has a data model and process model given by:

components' probability distributions. The other, which we prefer, is based on a sequence of conditional probabilities, as each random quantity is successively put into the conditioning set of random components. Moreover, the conditional approach allows (6.101) to be generalized to nonadditive (and non-Gaussian) decompositions of spatial, temporal, and spatio-temporal variability.

Example: Spatio-temporal Modeling of Temperatures in the Midwest

An example of the type of modeling motivated by (6.101) was given by Wikle, Berliner, and Cressie (1998), where the spatio-temporal process under study was the monthly average of maximum temperatures in a portion of the U.S. Midwest "corn belt" (centered around Iowa) for 240 months (i.e., 20 years from 1974 through 1993). Thus, space is continuous (over the corn belt) and time is discrete (monthly). From climatological knowledge, they specified the component μ to depend on space only (a Gaussian Markov random field with mean depending linearly on latitude and longitude; see Section 4.2.2 for more details on Markov random fields). The temporal trend γ was used to model an obvious seasonal cycle of temperatures (a linear combination of sine and cosine, each with period 12 months, where the two regression coefficients are themselves linear functions of latitude and longitude). The stochastic spatio-temporal term κ was modeled as being dynamical in time and descriptive in space, where the dynamics captured the month-to-month circulation changes in the atmosphere. This is clearly where a stochastic PDE or a stochastic difference equation, like those presented in Sections 6.3 and 7.2, could be used to model the smaller-scale (in both space and time) variability. Since large-scale weather patterns often exhibit fairly stationary longitudinal wavelike patterns in the mid-latitudes, and such patterns are fairly persistent in time with general west-to-east movement, it is possible that air temperatures at a given location in a given month could be related to air temperatures in regions to the west in the previous month. This suggested that a "lagged nearest neighbor" (Section 7.2.3) parameterization of the propagator matrix was reasonable. Specifically, the parameters corresponding to the neighbors to the east, west, north, and south were distinct, and the parameters corresponding to the same site at the previous time were allowed to vary with space.

Through (6.101), the spatio-temporal variability of temperature in and around Iowa was decomposed into a climatological component, a seasonal component, and a shorter-term dynamical component. An alternative, less satisfying decomposition would be to write the first part of (6.101), $\mu(\mathbf{s}; \boldsymbol{\theta}_t) + \gamma(t; \boldsymbol{\theta}_s)$, as a deterministic trend made up of a linear combination of (mixed) polynomials in space and time. Then, $\kappa(\mathbf{s}; t; \boldsymbol{\theta}_{s,t})$ would be a purely stochastic component with mean zero and covariance function given by, for example, one of the descriptive models presented in Section 6.1. As we mentioned above, there is clearly nonidentifiability associated with the spatio-temporal model (6.101): the extra dimension, time, does not allow one to avoid the difficulty discussed by Cressie (1993, p. 25): "What is one person's ... covariance structure may be another person's mean structure." Which decomposition to use should

depend as much as possible on the underlying scientific problem being studied. For example, the model of temperature in the Midwest given earlier, which incorporates climatology, seasonality, and changes in dynamical circulation, has far more explanatory power than a model with deterministic trend plus stationary stochastic error. Both may *describe* the particular spatio-temporal dataset well, but the latter model will generally do much worse in forecasting future temperatures in a spatial domain *similar* to (but different from) Iowa and surroundings. In conclusion, to the extent possible, we advocate choosing scientifically interpretable components of variation, including dynamical terms behaving (approximately) according to accepted physical relationships.

6.6.1 Dimension Reduction in a Dynamical Model

Recall the stochastic difference equation (6.70), which is a dynamical spatio-temporal statistical model. Wikle, Berliner, and Cressie (1998) use a model like this to capture the variability in temperature due to month-to-month circulation changes in the atmosphere. When space is discretized into pixels, $\{\kappa(s; t; \theta_{s,t}) : s \in D_s\}$ in (6.101) can be written as the vector κ_t , where the dimension of the vector is the number of pixels. The parameters $\{\theta_{s,t}\}$ are the elements of propagator matrices, $\{\mathbf{M}_t\}$, and the variance and covariance parameters in the error covariance Σ_η , as we now illustrate.

Let $D_t = \{0, 1, 2, \dots\}$, and assume that the spatio-temporal variability in (6.101) is given by the first-order linear dynamical process,

$$\kappa_t = \mathbf{M}_t \kappa_{t-1} + \eta_t, \quad t = 1, 2, \dots, \quad (6.102)$$

where $\{\mathbf{M}_t\}$ is a sequence of propagator matrices, $\{\eta_t\}$ is an *iid* sequence of error vectors each with mean $\mathbf{0}$ and covariance matrix Σ_η , and η_t is independent of κ_{t-1} .

When the number of pixels is very large, fitting the model (6.102) may not be possible without some form of *dimension reduction*. Dimension reduction can be achieved relative to the parameter space by using parameterizations suggested by “nearest neighbor,” PDE, or IDE models. Alternatively, dimension reduction can be relative to the process itself, for example, through a *spectral representation* (i.e., basis-function expansion),

$$\kappa_t = \Phi \alpha_t, \quad (6.103)$$

where Φ is a matrix of basis functions, and α_t is a vector of the associated spectral coefficients with *much* lower dimension (p_α , say) than that of κ_t . As described in Sections 4.1.4 and 7.1.3, we use the word “spectral” rather broadly to refer to general basis-function expansions, and thus there are many different choices available for these basis functions. For example, one might consider these to be orthogonal functions (e.g., Fourier, orthogonal polynomials such as Hermite polynomials, eigenvectors from a specified or estimated covariance

function), wavelets, splines, bisquares, discrete kernel convolutions, and so on, depending on the specific application. As an example, the spectral representation (6.103) could be based on truncating an infinite linear combination of a complete class of orthogonal basis functions. The truncation in such an expansion is a source of error that is often ignored; Wikle and Cressie (1999) handle it by assuming instead,

$$\kappa_t = \Phi \alpha_t + \xi_t, \quad (6.104)$$

where ξ_t is independent of α_t , and $\{\xi_t\}$ models all the hitherto-discarded, high-frequency terms through a relatively simple dependence structure, like $\xi_t \sim iid \text{Gau}(\mathbf{0}, \Sigma_\xi)$. In some cases, one might further decompose ξ_t into components associated with “small-scale” spatio-temporal processes and, perhaps, a white-noise process in space and time. In any case, a dynamical model can be fitted to $\{\alpha_t\}$, namely

$$\alpha_t = \mathbf{M}_{\alpha,t} \alpha_{t-1} + \gamma_t,$$

in a p_α -dimensional space (e.g., Wikle and Cressie, 1999).

If sufficient information is available, one can allow the basis functions to vary with time, perhaps on a different time-scale than the process of interest. That is, (6.103) could be generalized to

$$\kappa_t = \Phi_t \alpha_t, \quad (6.105)$$

where the columns of Φ_t are given basis functions that depend on t . Then define $\text{var}(\alpha_t) \equiv \Sigma_{\alpha,t}$, whose entries are considered to be a subset of the parameters $\{\theta_{s,t}\}$. In the case where these basis functions are estimated, there are potential identifiability concerns with modeling the dynamics of α_t , and hence simple dynamical structures (e.g., random walks) for α_t need to be specified.

If possible, one should account for the truncation associated with the expansion (6.105), analogously to the model given in (6.104):

$$\kappa_t = \Phi_t \alpha_t + \xi_t, \quad t = 0, 1, \dots, \quad (6.106)$$

where ξ_t is independent of α_t , and assume $\xi_t \sim iid \text{Gau}(\mathbf{0}, \Sigma_\xi)$. Once again, $\{\xi_t\}$ plays the role of modeling residual spatio-temporal or spatial dependence (Section 6.4.2). In some cases, this residual term may just account for variation, but without spatial or temporal dependence; for example, Cressie, Shi, and Kang (2010) use (6.106) with $\Sigma_\xi = \sigma_\xi^2 \mathbf{I}$, to capture pixel-scale variability in a model applied to a remote-sensing problem, or Σ_ξ could be diagonal.

If one has sufficient knowledge to specify the time-varying basis functions, then it is possible to consider more complicated dependence in the process α_t . That is, one could consider a first-order dynamical process such as

$$\begin{aligned} \kappa_t &= \Phi_t \alpha_t + \xi_t, \\ \alpha_t &= \mathbf{M}_{\alpha,t} \alpha_{t-1} + \gamma_t, \end{aligned} \quad (6.107)$$

where $\mathbf{M}_{\alpha,t}$ is a $p_\alpha \times p_\alpha$ propagator matrix, and $\boldsymbol{\gamma}_t$ is a p_α -dimensional vector independent of $\boldsymbol{\alpha}_{t-1}$ [e.g., see Cressie, Shi, and Kang (2010)]. Such a model implies marginal covariances for $\boldsymbol{\kappa}_t$ that are nonstationary and nonseparable (Sections 6.1.2 and 6.1.3). An example using (6.107) in an EHM is given in Section 9.2.

6.6.2 Hierarchical Spatio-Temporal Statistical Modeling

We close this section with an indication of what is to come in Chapter 7. In Chapter 6, we have made the case for preferring dynamical (conditional-probability-based) spatio-temporal models over descriptive (marginal-probability-based) spatio-temporal models. However, we have not yet shown how the dynamical models fit into a hierarchical statistical structure. Consider observations and potential observations described by $\{Z(\mathbf{s}; t) : \mathbf{s} \in D_s, t \in D_t\}$, or $Z(\cdot; \cdot)$. We are careful to distinguish $Z(\cdot; \cdot)$ from $\{Y(\mathbf{s}; t) : \mathbf{s} \in D_s, t \in D_t\}$, or $Y(\cdot; \cdot)$, which is the physical/chemical/biological/economic/etc. process of interest.

We observe $Y(\cdot; \cdot)$ imperfectly because the observations are incomplete and noisy. We write the data $\mathbf{Z}$ in a way that emphasizes their spatio-temporal coordinates:

$$\mathbf{Z} \equiv (Z(\mathbf{s}_i; t_i) : i = 1, \dots, m)', \quad (6.108)$$

and we often assume that the error due to measurement is additive, namely

$$Z(\mathbf{s}_i; t_i) = Y(\mathbf{s}_i; t_i) + \varepsilon(\mathbf{s}_i; t_i), \quad i = 1, \dots, m, \quad (6.109)$$

where $\varepsilon(\mathbf{s}_i; t_i)$ represents a random measurement-error component, whose probability distribution is specified. For example, if $\boldsymbol{\varepsilon} \equiv (\varepsilon(\mathbf{s}_1; t_1), \dots, \varepsilon(\mathbf{s}_m; t_m))'$, then suppose that $\boldsymbol{\varepsilon} \sim \text{Gau}(\boldsymbol{\mu}_\varepsilon, \boldsymbol{\Sigma}_\varepsilon)$; often, $\boldsymbol{\mu}_\varepsilon = \mathbf{0}$ and $\boldsymbol{\Sigma}_\varepsilon = \sigma_\varepsilon^2 \mathbf{I}$, because the measuring instrument performs unbiasedly, consistently, and independently at each place and time a measurement is taken. The best instruments are built so that $\boldsymbol{\mu}_\varepsilon = \mathbf{0}$ and σ_ε^2 is as small as possible. The assumption (6.109), of additive Gaussian measurement error, is an example of a *data model*. We can generalize the data model by expressing it in terms of conditional probabilities:

$$[\mathbf{Z} | Y(\cdot; \cdot), \boldsymbol{\theta}_D], \quad (6.110)$$

where $\boldsymbol{\theta}_D$ are *parameters* in the data model. This is the *first* level of the HM.

The *second* level of the HM is the *process model*:

$$[Y(\cdot; \cdot) | \boldsymbol{\theta}_P], \quad (6.111)$$

where $\boldsymbol{\theta}_P$ are *parameters* of the process model. In most of the sections of this chapter, we have concentrated on the distribution of the hidden process,

(6.111) and various ways were given to (partially) specify it, either dynamically or descriptively.

Bayes' Theorem tells us that the posterior distribution of $Y(\cdot; \cdot)$ given the data (and the parameters), is

$$[Y(\cdot; \cdot) | \mathbf{Z}, \theta] \propto [\mathbf{Z} | Y(\cdot; \cdot), \theta_D][Y(\cdot; \cdot) | \theta_P], \quad (6.112)$$

where $\theta \equiv \{\theta_D, \theta_P\}$. From (6.112), we can predict (parts of, or functionals of) the “hidden” process $Y(\cdot; \cdot)$. Notice that the right-hand side of (6.112) is simply the product of the data model (6.110) and the process model (6.111). What should be done about the unknown parameters θ ?

The *empirical hierarchical model* (EHM) performs separate estimation of θ , based on the data $\mathbf{Z}$ or other data; the resulting $\hat{\theta}$ is “plugged into” the posterior distribution given by (6.112); see Section 9.2 for an example. The *Bayesian hierarchical model* (BHM) posits a *third* level of the hierarchy, namely the *parameter model* (or prior),

$$[\theta] = [\theta_D, \theta_P]. \quad (6.113)$$

Then, Bayes' Theorem allows inference on *all* unknowns, $Y(\cdot; \cdot)$ and θ , in the same manner through the posterior distribution:

$$[Y(\cdot; \cdot), \theta | \mathbf{Z}] \propto [\mathbf{Z} | Y(\cdot; \cdot), \theta_D][Y(\cdot; \cdot) | \theta_P][\theta], \quad (6.114)$$

where now the right-hand side of (6.114) is simply the product of the distributions specified at the first level (data model (6.110)), the second level (process model (6.111)), and the third level (parameter model (6.113)) of the hierarchy. See Sections 9.1, 9.3, and 9.4 for examples. A spatio-temporal HM and the inferences associated with it might seem like a way to do *data assimilation*. In fact, it is (Wikle and Berliner, 2007).

Now, when we think of spatio-temporal statistical models as time series of spatial processes, we can think of optimal smoothing, filtering, and forecasting of the processes. By allowing the (discretized) spatial dimension to define a vector of attributes, $\mathbf{Y}_t$, we see that spatio-temporal prediction (smoothing, filtering, forecasting) can be obtained from a multivariate version of optimal prediction (e.g., Shumway and Stoffer, 2006). As has already been noted, Chapter 9 considers both the EHM and the BHM approaches to spatio-temporal statistical analysis of various real-world phenomena, where dynamical spatio-temporal models are featured.

6.7 BIBLIOGRAPHIC NOTES

General References to Spatio-Temporal Statistics

Books and edited volumes that have contributed to the subject's recent evolution are Le and Zidek (2006), Finkenstädt, Held, and Isham (2007), and Gelfand

et al. (2010, Part V). That literature has developed largely from spatial statistics, although Bennett (1979) develops models from time series. Banerjee, Carlin, and Gelfand (2004) and Schabenberger and Gotway (2005) each have a chapter devoted to spatio-temporal modeling, and Cressie (1991, 1993) has a small discussion of spatio-temporal statistics in each of the three parts of his book. Higdon (2007) gives a primer on space-time modeling from a Bayesian perspective; Chen, Fuentes, and Davis (2006) review the literature relevant to the study of environmental processes; Christakos (2000) presents a rather specialized approach from a geostatistical viewpoint; and Renshaw (1991) gives a stochastic processes approach to modeling biological populations in space and time.

Models for Spatio-Temporal Covariance Functions

There has been a little work done on models for spatio-temporal variograms (Fernandez-Casal, González-Manteiga, and Febrero-Bande, 2003). Generally the literature has concentrated on spatio-temporal covariance functions; Sherman (2011) reviews models for spatio-temporal covariance functions. Spectral representations of stationary spatio-temporal covariance functions are another way to describe the spatio-temporal dependence; see the review by Fuentes, Guttorp, and Sampson (2007).

Separable spatio-temporal models were used in early studies of spatio-temporal statistics; their convenience (positive-definiteness and efficient computations) outweighed their obvious lack of flexibility. Since 1999, there has been substantial work done to understand separability, to understand full symmetry, and to construct quite general classes of nonseparable spatio-temporal covariance functions (Cressie and Huang, 1999; Kyriakidis and Journel, 1999; De Iaco, Myers, and Posa, 2002; Gneiting, 2002; Stein, 2005; Genton, 2007; Gneiting, Genton, and Guttorp, 2007; and Jun and Stein, 2007). Mathematically speaking, one seeks functions of spatial lag $\mathbf{h}$ and temporal lag τ that are non-negative definite and *not* of product form. There have been some ingenious constructions, although only a few of them have been fitted to spatio-temporal data (De Cesare, Myers, and Posa, 2001; Ma, 2003, 2005; Stein, 2005; Porcu, Gregori, and Mateu, 2006; Porcu, Mateu, and Saura, 2008). Nonseparable, nonstationary, spatio-temporal covariance functions have been considered by Fuentes, Chen and Davis (2008) and Cressie, Shi, and Kang (2010).

Spatio-Temporal Kriging

Kriging typically involves the inverse of the data covariance matrix. A general difficulty with inversion of this matrix has led to simplifying assumptions about spatio-temporal covariance functions, such as separability and additivity. Separability is discussed in the previous paragraph. A model that adds a purely spatial covariance to a purely temporal covariance was considered by Bilonick (1985) and Rouhani and Hall (1989), although for certain data configurations the covariance matrix can be singular. One could use spatio-temporal neighborhoods to reduce the order of the kriging equations (Haas, 1995, 2002). Models for spatio-temporal kriging do not have to be

dynamical (e.g., Cressie and Majure, 1997; Hartfield and Gunst, 2003; Huang and Hsu, 2004; Lophaven, Carstensen, and Rootzen, 2004; Augustin et al., 2009; Bodnar and Schmid, 2010).

Stochastic Partial Differential Equations

A physical system defined by a stochastic PDE in space and time implies a spatio-temporal covariance function. Whittle (1963, 1986) has been a major contributor to this area; he is also acknowledged by Heine as having suggested the problem solved in the seminal paper of Heine (1955).

Spatio-Temporal Models Based on Conditional Probabilities

Decomposition of complex spatio-temporal processes can often be decomposed into relatively simple conditional-probability-based models. Early examples of this include the articles by Handcock and Wallis (1994), Brown, Le, and Zidek (1994), Waller et al. (1997), Knorr-Held and Besag (1998), and Wikle, Berliner, and Cressie (1998). It is natural to assume that the immediate-past state of the process affects the current state (e.g., Guttorp, Meiring, and Sampson, 1994; Huang and Cressie, 1996; Mardia et al., 1998; Wikle and Cressie, 1999; Wikle et al., 2001; Stroud, Müller, and Sansó, 2001). Brown et al. (2000) discuss stochastic PDEs and their discretization, as do Wikle (2003b), Xu and Wikle (2007), Wikle and Hooten (2006), and Hooten and Wikle (2008). Stochastic IDEs were proposed by Wikle and Cressie (1999), Storvik, Frigessi, and Hirst (2002), Wikle (2002a), and Xu, Wikle, and Fox (2005).

Times Series of Spatial Processes

Before we review this area, we would like to point to an alternative approach to modeling spatio-temporal data, namely as a spatial process of time series (Bennett, 1979; Rouhani and Wackernagel, 1990; Sansó, Schmidt, and Nobre, 2008).

When considering time series of spatial processes, the choice is generally between time series of geostatistical processes or time series of lattice processes. Many spatio-temporal studies address problems in the environmental sciences, for which geostatistical processes evolving in time are usually chosen (e.g., Haslett et al., 2006; Lemos and Sansó, 2009; and Hering and Genton, 2010). These are the predominant class of models studied in Chapter 7, and hence for these we refer the reader to the Bibliographic Notes in Section 7.8. Nevertheless, there are two reviews of the field that should be mentioned here, namely Kyriakidis and Journel (1999) and Chen, Fuentes, and Davis (2006).

For studies in the social sciences in general, and for Public Health studies in particular, spatio-temporal data are usually aggregated spatially to small areas (often for confidentiality reasons). Hence, lattice processes evolving in time are usually chosen [Pavía, Larraz, and Montero (2008), is an exception to this]. Examples include Waller et al. (1997), Knorr-Held and Besag (1998), Knorr-Held (2000), Lagazio, Dreassi, and Biggeri (2001), Assunção, Reis, and Oliveira (2001), Mugglin, Cressie, and Gemmell (2002), Carlin and Banerjee

(2003), and Norton and Niu (2009). However, there are studies in the environmental sciences that have also used lattice processes (particularly Markov random fields) evolving in time to capture dependence on discretized space (e.g., Rathbun and Cressie, 1994; Lavine and Lozier, 1999; Allcroft and Glasbey, 2003; Zhu, Huang, and Wu, 2005; Daniels, Zhou, and Zou, 2006; Zhang and Zhu, 2008; Zhu et al., 2008; Ugarte, Goicoa, and Militino, 2010).

Spatio-Temporal Point Processes

A number of references were already given in Section 6.5. The use of an HM is not all that common in the literature, although obviously a Cox process can be thought of this way. Much of the modeling and inference has been based on a model that we express here generically as $[Z|\theta]$. Even for the Cox process, marginalization has often been carried out through, $[Z|\theta] = \int [Z|Y][Y|\theta] dY$, where $\lambda \equiv e^Y$ (for example) is the intensity of the inhomogeneous Poisson process, $Z|Y$. Estimation of θ through maximum likelihood (Veen and Schoenberg, 2008) or “pseudo” likelihood (Schoenberg, 2005; Baddeley and Turner, 2006; Diggle, 2006) are the predominant methods proposed. There has been some Bayesian inference (e.g., Brix and Diggle, 2001; Brix and Møller, 2001; Diggle, Rowlingson, and Su, 2005) in the literature, and it is clear that this will be an area of growth in the future.

Identifying clusters in spatio-temporal point patterns is important for disease surveillance and defending against bio-terrorism. This has attracted considerable interest since 2001; for example, see Clark and Lawson (2002), Diggle, Rowlingson, and Su (2005), Kulldorf et al. (2005), and Assunção and Correa (2009).

In the area of spatio-temporal marked point processes, there have been innovative contributions to modeling atmospheric precipitation through stochastic rainfall models (e.g., see Gupta and Waymire, 1979; Smith and Karr, 1985; Rodriguez-Iturbe and Eagleson, 1987; Velghe et al., 1994). When the mark process is based on random sets, stochastic models of biological growth can be built that are very flexible and interpretable (e.g., Cressie and Hulting, 1992; Chiu and Quine, 2001; Jensen et al., 2007; and the sequence of random-set models presented in Figures 4.19b, 4.20b, and 4.20d).

